

Deep Learning for Computer Vision

Lecture 1: Introduction and Overview

Constantin Pape

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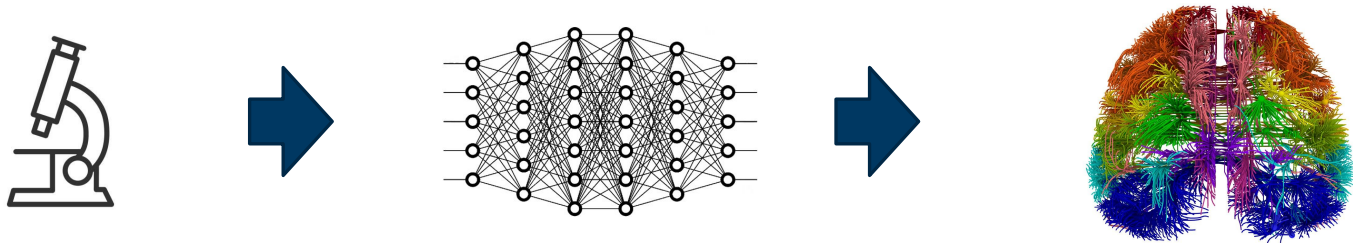
@cppape.bsky.social

<https://user.informatik.uni-goettingen.de/~pape41/>

Background & Organization

Introduction

Deep Learning for Microscopy



Large Microscopy Data analyzed with **deep learning methods** for better **understanding of biology**

- Juniorprofessor at IfI and CIDAS
- Group “Computational Cell Analytics”

Introduction

Contents of this course:

- Deep learning basics
 - Learning algorithms
 - Network architectures
- Applications in computer vision
 - Common tasks: Classification, segmentation, detection
 - Image synthesis & generative models
 - Vision foundation models
 - ...
- Not covered in detail: applications in other domains, e.g. NLP
 - Methodology in these domains is similar

Introduction

Based primarily on **Deep Learning** by Alexander Ecker
(*same module; B.Inf.1237*)

Differences:

- No natural language processing
- More focus on computer vision applications
 - More details on segmentation & object detection
 - Vision transformers
 - Video processing, point clouds & neural rendering

If you have heard Deep Learning already and have qualified for the exam:
Don't need to submit exercises (more later)

Related Courses - Lectures

Prerequisite
Knowledge

1. Summer term: **Machine Learning** (*B.Inf.1236*)
2. Winter term: **Deep Learning for Computer Vision** (*this course; B.Inf.1237*)
3. Summer term: **Deep Learning for Natural Language processing** (*M.Inf. 2202*)
4. Winter term: **Language as Data** (*B.Inf.1248*)
5. Winter term: **Probabilistic Machine Learning** (*M.Inf.2201*)
6. Winter term: **Graph Machine Learning**
https://eckerlab.org/lectures_and_courses/graph_learning/

Related Courses - Seminars

Seminars & Practical courses

1. Winter term: **Current Topics in Machine Learning** (*M.Inf.2241*)
2. Summer term: **Deep Learning in Biology and Medicine** (*M.Inf.2244*)
3. Winter Term: **Going Deeper into Computer Vision**
 - Advanced Course that can be heard **in addition** to this lecture.
 - Implementing deep learning architectures from the ground up.
 - By Dr. Timo Lueddecke.
 - **More details later.**

My seminar

Course topics & schedule

Overview and introduction (today)

Neural Network Recap (Lecture 2)

Convolutional neural networks (CNNs) (Lecture 3)

Vision tasks & Transfer learning (Lecture 4)

Segmentation & Image to image tasks (Lecture 5)

Object detection & Instance segmentation (Lecture 6)

Visualizing CNNs & Perceptual loss functions (Lecture 7)

Generative models for image synthesis (Lecture 8)

(Vision) Transformers (Lecture 9)

Weak supervision, Domain adaptation & Self-supervision (Lecture 10)

Vision Foundation Models (Lecture 11)

Videos & point clouds (Lecture 12)

Diffusion Models / Flow Matching & Neural rendering (Lecture 13)

Summary + Applications in biology and medicine (Lecture 14)

Additional literature

Primary book:

- Goodfellow, Bengio, Courville:
Deep Learning
<https://www.deeplearningbook.org>

Other resources:

- Aston Zhang, Zachary C Lipton, Mu Li and Alexander J Smola:
Dive into Deep Learning
<http://d2l.ai/>
- Eli Stevens:
Deep Learning with PyTorch
<https://pytorch.org/assets/deep-learning/Deep-Learning-with-PyTorch.pdf>

Prerequisite knowledge

Generally:

- **Scientific python (numpy!) / jupyter**
- Linear algebra
- Basic probability theory
- Machine Learning (e.g. B.Inf.1236)

B.Sc. Applied Data Science / Applied Computer Science:

- B.Inf.1131 – Data Science I: Algorithmen und Prozesse
- B.Inf.1236 – Machine Learning
- Recommended for 5th+ semester

M.Sc. Applied Computer Science:

- M.Inf.1151 – Data Science und Big Data Analytics
- B.Inf.1236 – Machine Learning

Prerequisite knowledge

Recap lectures

- Recap 1: Math Basics
- Recap 2: ML & DL Basics
- Recap 3: Computer Vision Basics

Recordings from last years are available on Stud.IP

Vorlesung mit Übung: Deep Learning for Computer Vision (WiSe 2025/26)

Verwaltung Übersicht Forum Teilnehmende Dateien Ablaufplan General Information Wiki Blubber

 Dateien

Aktionen

-  Ordner bearbeiten
-  Neuer Ordner
-  Dokument hinzufügen

Dateien hochladen



Neue Dateien zum Hinzufügen per

 / lecture_recordings

Ein Ordner für Materialien, welche nur zum Download zu Verfügung gestellt werden sollen.
Den Inhalt des Ordners können nur Lehrende und TutorInnen verändern.

Lecture Recordings

☐	Typ	Name ^	Größe	Autor/-in
☐		recap_01_math_basics.mp4	140.9 MB	Pape, Const
☐		recap_02_mlanddl.mp4	202.9 MB	Pape, Const
☐		recap_03_computer_vision.mp4	132.1 MB	Pape, Const

Organization - General

**No emails or
messages on
Stud.IP please!**

Communication

- Matrix: <https://chat.uni-goettingen.de>
 - Join the rom #DL-For-CV-2526
 - Direct communication with organisers & tutors (see handles on next slide)
 - Tutors will share links for individual channels later.

Lectures

- Tuesday 10:15-12:00 (MN14)
- Slides will be made available before, recordings after the lecture
- Please come in person if possible

Organization - Exercises

Exercises

- 6 homework assignments. Two weeks to work on them.
- Groups of 3 students
- Present solution to your tutor in tutorials, tutor will decide if pass or fail
 - Pass: ca. 80% are solved correctly
 - If you fail: you can correct your solution and submit again once
 - Need to pass 5/6 to qualify for the exam

PhD Students

4 credits for attendance. 6 for also doing the exercises.
You don't need to write the exam.

Issues & help with the exercises

- Ask in the chat
- Ask your tutor during office hours

Organization - Tutorials

Tutorials

- Biweekly, first tutorials week of Nov. 10th
- Tutors will offer office hours in weeks without tutorial

Schedule

- Tuesday 16:00-18:00 (Informatik-Provisorium 0.103)
- Wednesday 14:00-16:00 (Informatik-Provisorium 0.103)
- Wednesday 16:00-18:00 (Informatik-Provisorium 0.103)
- Thursday 10:00-12:00 (Informatik-Provisorium 0.103)
- Thursday 14:00-16:00 (Zoom)
- Friday 10:00-12:00 (Informatik-Provisorium 0.103)
- Friday 14:00-16:00 (Zoom)

Organization - Team

- Constantin Pape @pape41:uni-goettingen.de
 - Lecturer
- Anwai Archit @archit:uni-goettingen.de
 - Organization
- Sushmita Nair @nair3:uni-goettingen.de
 - Organization
- Carolin Teuber @carolin.teuber:uni-goettingen.de
 - Tutorial: Tuesday 16-18
- Sarah Muth @sarah.muth:uni-goettingen.de
 - Tutorial: Wednesday 14-16
- Oleg Bakumenko: @obakume:umg.eu
 - Tutorial: Wednesday 16-18
- Marei Freitag @marei.freitag:uni-goettingen.de
 - Tutorial: Thursday 10-12
- Azhar Akhmetova @azhar.akhmetova:uni-goettingen.de
 - Tutorial: Thursday 14-16 (**Zoom**)
- Benjamin Eckhardt @benjamin.eckhardt:uni-goettingen.de
 - Tutorial: Friday 10-12
- Luca Freckmann @freckmann15:uni-goettingen.de
 - Tutorial: Thursday 14-16 (**Zoom**)

Organization - Tutorials

Signing up for the tutorials

Find a group of 3, sign up via the google form: <https://forms.gle/wKyjNFi2JyAiBWGL7>

- Enter group name (arbitrary, you can use surnames)
- Enter at least 3 priorities for tutorials
- Enter the group member details

Can't find a group? Ask in the chat room!

We will assign the tutorials next Monday (Nov 3)

If you haven't found a group by then: we will have another form to build groups.

But please try to find a group now!

The screenshot shows a Google Form titled "Tutorial: Deep Learning for Computer Vision (Groups)". It includes a legend indicating that an asterisk (*) denotes a required question. The form has three main sections: 1. "Group Name" with a text input field labeled "Your answer". 2. "Priority 1" with a list of seven time slots, each preceded by a radio button. The slots are: Tuesday 04:00 pm - 06:00 pm, Wednesday 02:00 pm - 04:00 pm, Wednesday 04:00 pm - 06:00 pm, Thursday 10:00 am - 12:00 am, Thursday 02:00 pm - 04:00 pm (via Zoom), Friday 10:00 am - 12:00 am, and Friday 02:00 pm - 04:00 pm (via Zoom). 3. "Priority 2" with a single visible radio button for Tuesday 04:00 pm - 06:00 pm.

Tutorial: Deep Learning for Computer Vision (Groups)

* Indicates required question

Group Name *

Your answer

Priority 1 *

☐ Tuesday 04:00 pm - 06:00 pm

☐ Wednesday 02:00 pm - 04:00 pm

☐ Wednesday 04:00 pm - 06:00 pm

☐ Thursday 10:00 am - 12:00 am

☐ Thursday 02:00 pm - 04:00 pm (via Zoom)

☐ Friday 10:00 am - 12:00 am

☐ Friday 02:00 pm - 04:00 pm (via Zoom)

Priority 2 *

☐ Tuesday 04:00 pm - 06:00 pm

Organization - Tutorials

Presenting the exercises

- Ratio of students (>150 signed up) to tutors (7) is quite high.
 - We need to be strict with evaluation of the exercises.
- You will be assigned a timeslot for presenting your exercise in the tutorial.
 - The tutors will contact you about the timeslots.
- **Prepare** for showing the exercise to your tutor as a group
 - Present by going through your notebook
 - Present the results, discuss difficulties, unexpected results, questions
 - only discuss code in detail if there is a good reason (e.g. difficult part)
 - don't present optional parts in detail
 - Decide which part is presented by which group member **beforehand and prepare your part.**
 - Present the whole exercise in **15 minutes or less.**
 - Each group member should present a part.

Organization - Exercises

Technical

- Python 3 + **PyTorch** (+ numpy, matplotlib, etc.)
- Jupyter Notebooks
- You will need a GPU for the exercises
- Recommended: GWDG HPC Jupyter:
 - Instructions at:
https://docs.google.com/document/d/1frgBvU9ZRcR7PkMqOuK5eTj5RphZBmjz_i6OgA1DVwE/edit?usp=sharing

Organization - Exercises

Technical

- Python 3 + **PyTorch** (+ numpy, matplotlib, etc.)
- Jupyter Notebooks
- You will need a GPU for the exercises
- Recommended: GWDG HPC Jupyter:
 - **You will get an invite to this via your student mail.**

Hello [redacted]

you have been added to the HPC project "Lecture: Deep Learning for Computer Vision WS 25/26" with the username u[redacted] the system "SCC".

Please click on the following link to get more information about your project: [go to project](#).

Alternatively, copy-paste the following plain-text URL into your browser: [https://hpcproject.gwdg.de/login/?next=\[redacted\]](https://hpcproject.gwdg.de/login/?next=[redacted])

You can login with the account u[redacted] the HPC login nodes SCC:

- login-mdc.hpc.gwdg.de
- glogin-p3.hpc.gwdg.de
- glogin-p2.hpc.gwdg.de
- glogin-gpu.hpc.gwdg.de

Instructions can be found on the [documentation](#) page.

Yours sincerely,

GWDG HPC Project Management System

Organization - Exercises

Technical

- Python 3 + **PyTorch** (+ numpy, matplotlib, etc.)
- Jupyter Notebooks
- You will need a GPU for the exercises
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 - Instructions at:
https://docs.google.com/document/d/1frgBvU9ZRcR7PkMqOuK5eTj5RphZBmjz_i6OgA1DVwE/edit?usp=sharing

Alternative: Kaggle Notebooks

See instructions at:

<https://gist.github.com/anwai98/444d3ff11c1365f2c610579d5df0830b>

Organization - Exercises

submission

- Will share the exercise Tuesday after the lecture on StudIP
 - **Stud.IP -> Files -> Homework**
- Submission
 - Upload via Stud.IP (**Stud.IP -> Files -> Submission of Homework x -> Tutorial y**)
 - Upload one solution per group, only the notebook file:
<surname1>_<surname2>_<surname3>.ipynb
 - Submission Deadline Tuesday at 10:00 (before the start of the lecture)

First exercise will be available soon!

Organization - Schedule

- Ex.1: 28.10. - 11.11., (Tutorials: Week of 10.11.)
- Ex.2: 11.11. - 25.11., (Tutorials: Week of 24.11.)
- Ex.3: 25.11. - 09.12., (Tutorials: Week of 08.12.)
- Ex.4: 09.12. - 07.01., (Tutorials: Week of 06.01.)
- Ex.5: 07.01. - 21.01., (Tutorials: Week of 20.01.)
- Ex.6: 21.01. - 04.02., (Tutorials: Week of 04.02.)

There will be a sample exam published with exercise 6 and discussed in the last lecture.

Organization - Exam / Credits

- Lecture gives 6 credits
- To pass: written exam
 - First exam: 17.02.2025, 10:00-12:00 (MN08)
 - Second exam: 07.04.2025, 10:00-12:00 (MN08)
- Grade will be determined by the exam
 - Qualification: pass 5 / 6 exercise

Organization - Exam / Credits

- Lecture gives 6 credits
- To pass: written exam
 - First exam: 17.02.2025, 10:00-12:00 (MN08)
 - Second exam: 07.04.2025, 10:00-12:00 (MN08)
- Grade will be determined by the exam
 - Qualification: pass 5 / 6 exercise
- If you have already heard **Deep Learning for Computer Vision** in prev. semesters and have exam qualification: you don't need to do exercises again!

Recommendation: Follow the lecture!

Organization - Exam / Credits

- Lecture gives 6 credits

- If you're a PhD Student and want credits for the lecture:
 - 4 credits for attendance (need to attend at least 10 lectures in person, sign attendance after the lecture)
 - 6 Credits if you do the exercises (no need for attendance check)
 - No need to write exam
- If you're not a PhD Student and have exam qualification: you need to do exercises again!

Recommendation: Follow the lecture!

Organization - Exam / Credits

- Lecture gives 6 credits

- If you're a PhD Student and want credits for the lecture:
 - 4 credits for attendance (need to attend at least 10 lectures in person, sign attendance after the lecture)
 - 6 Credits if you do the exercises (no need for attendance check)
 - No need to write exam

you need to do exercises as

Recommendation: Follow the lecture!

You also have to sign up for the exercises (and exam) in FlexNow.
Deadline: 12.12.25
Important: No extensions to this deadline! Sign up as soon as possible.

Organization - Advanced course

Going Deeper into Computer Vision by Dr. Timo Lueddecke.

- Can be heard **in addition** to this lecture for 4 extra credits.
- In-depth discussion of DL architectures and implementation from the ground up.
 - 3 lectures and 3 exercises during the semester.
- Limited to ca. 20 participants.
- First lecture / organization:
Wednesday 05.11., 14:00-16:00 in Provisorium 0.103
 - If you are interested in attending go there
- More information and preliminary sign-up:
https://studip.uni-goettingen.de/dispatch.php/course/details?sem_id=fb98a146b300e16f1817ea6cfaa28725&again=yes

Organization - Next steps

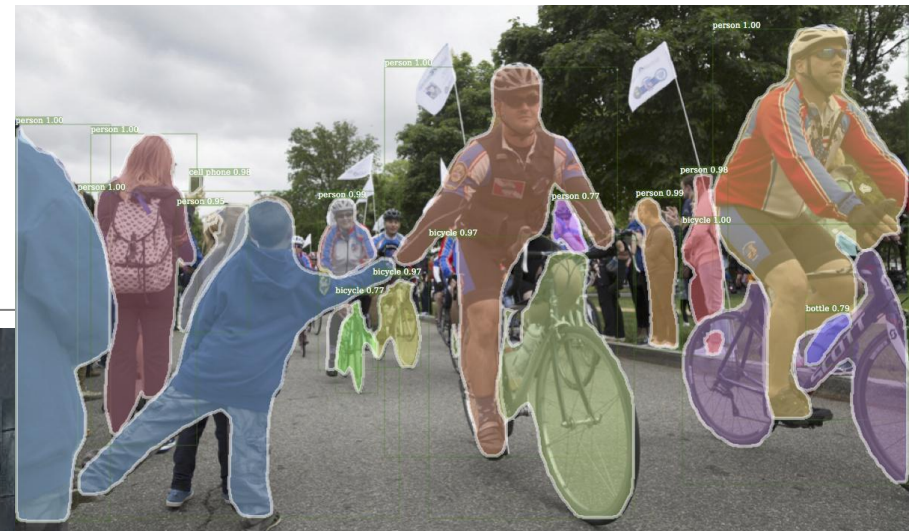
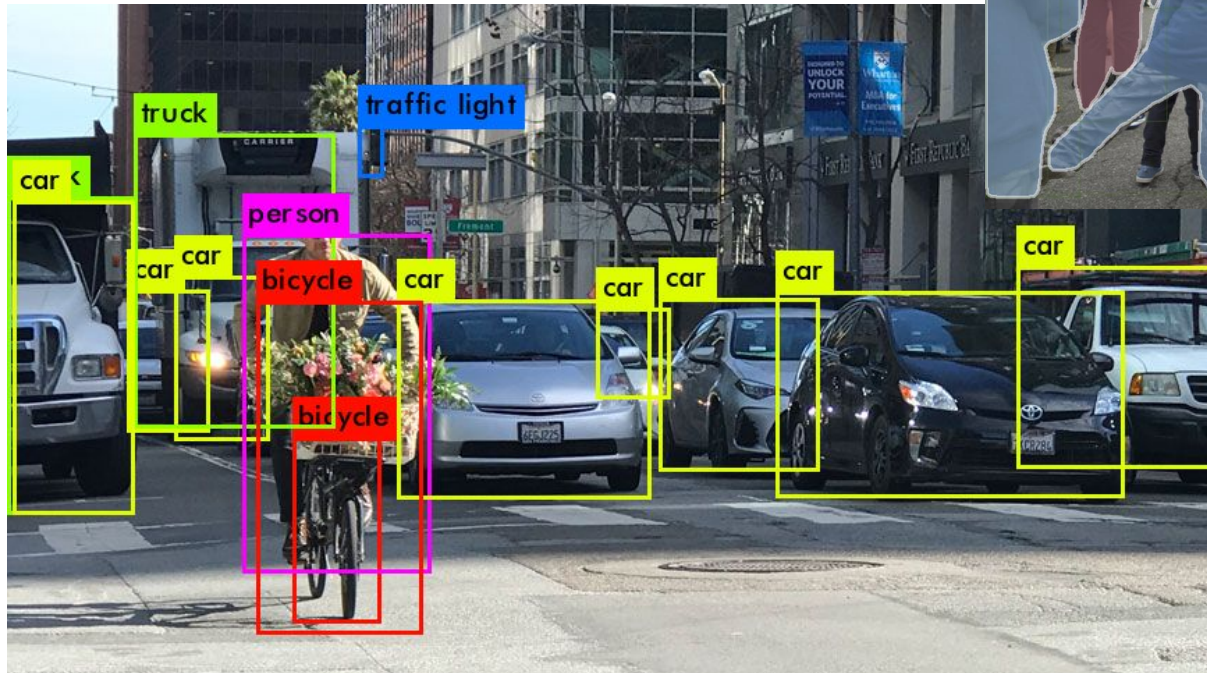
- Join #DL-For-CV-2526 on <https://chat.uni-goettingen.de> for communication and announcements
- Form a group of three, sign up for tutorials at <https://forms.gle/wKyjNFi2JyAiBWGL7>
- First exercise will be available after the lecture, solve it with your group and submit by Nov. 11 via Stud.IP
- Present the solution with your group in the first tutorials
- Sign up for exercises on FlexNow as soon as possible, deadline: 12.12.25
- Check out recap lectures if you are unsure about your prior knowledge

Questions?

The AI revolution

2012 – PRESENT

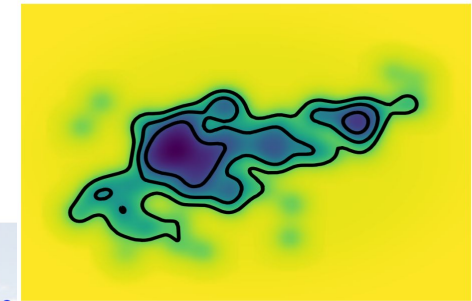
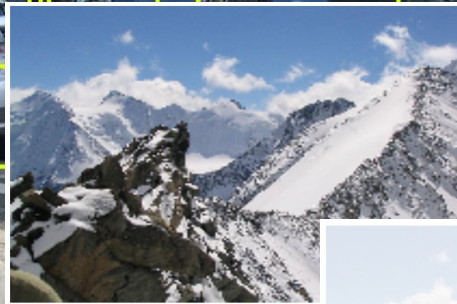
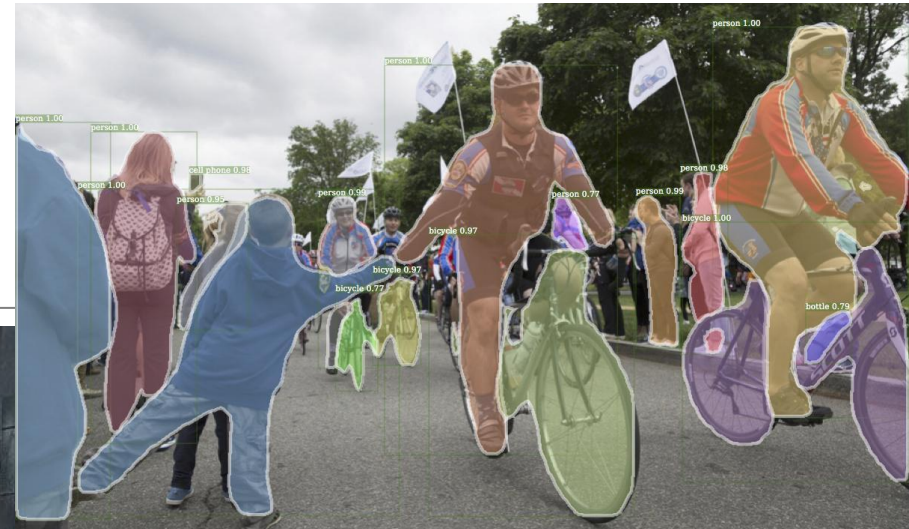
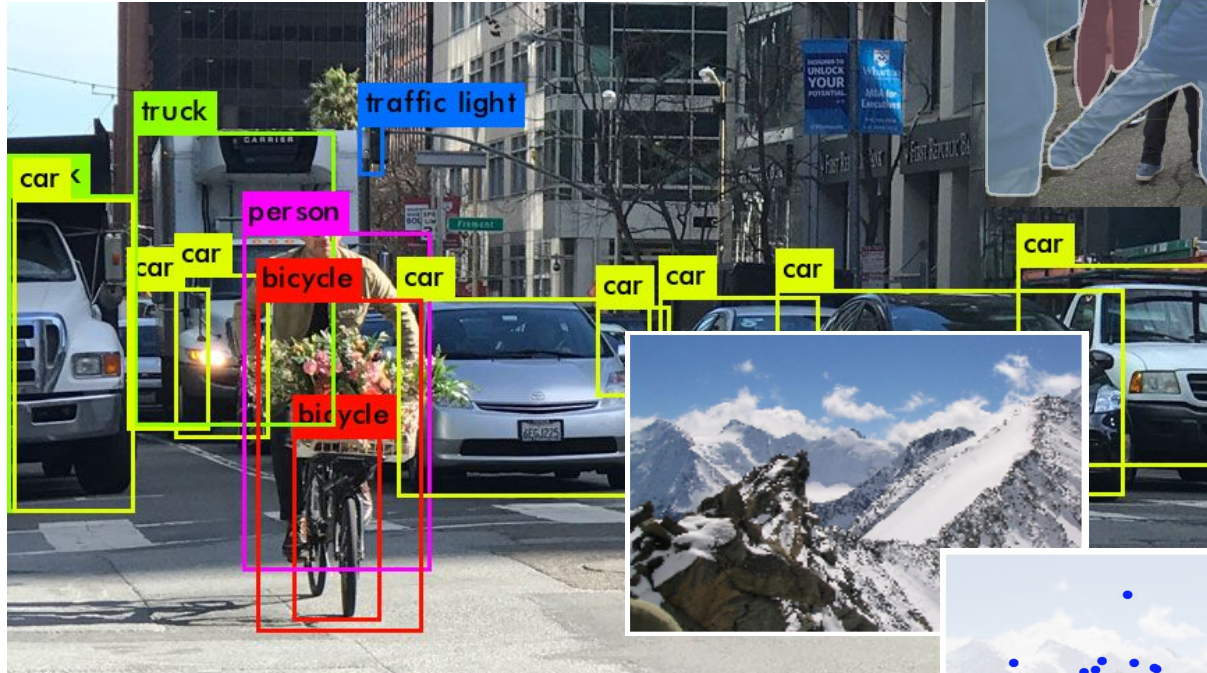
Computer Vision



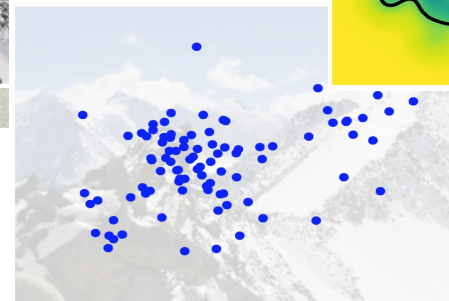
<https://github.com/facebookresearch/Detectron>

https://medium.com/@jonathan_hui/real-time-object-detection-with-yolo-yolov2-28b1b93e2088

Computer Vision



Eye
fixations



Saliency
map

Kümmerer, Theis, Bethge, 2015, ICLR Workshop
<https://github.com/facebookresearch/Detectron>

https://medium.com/@jonathan_hui/real-time-object-detection-with-yolo-yolov2-28b1b93e2088

Image generation

DALL·E 2 can create images from a caption.



An astronaut riding a horse in a photorealistic style.

Teddy bears mixing sparkling chemicals as mad scientists as a 1990s Saturday morning cartoon.



Language processing



Translator

Linguee

DeepL Pro

Blog

Info



Translate from **German** (detected) ▾

Die Europäische Zentralbank hat das Ende ihrer Anleihenkäufe beschlossen. Nur noch bis zum Jahresende will die Notenbank zusätzliche Milliarden in Wertpapiere von Staaten und Unternehmen stecken, wie die EZB am Donnerstag in Frankfurt mitteilte. Anschließend lässt sie das Programm auslaufen.



📎 Translate document

Translate into **English** ▾

The European Central Bank has decided to end its bond purchases. Only until the end of the year does the central bank intend to invest additional billions in securities of states and companies, as the ECB announced in Frankfurt on Thursday. It will then phase out the programme.



Large language models / ChatGPT

✦ GPT-4



Summarize the topics for an introductory deep learning lecture in five concise bullet points.



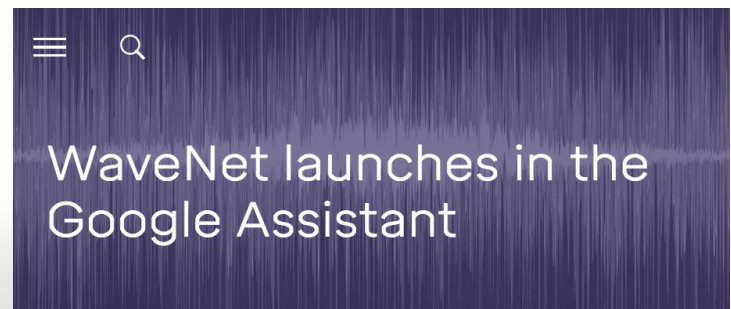
Certainly! Here's a concise summary for an introductory deep learning lecture:



1. **Introduction to Neural Networks:** Basics of what neural networks are, their architecture (e.g., neurons, layers, and weights), and a historical perspective.
2. **Activation Functions & Forward Propagation:** Explanation of activation functions (e.g., ReLU, Sigmoid, and Tanh) and how data flows through a network (i.e., the process of forward propagation).
3. **Loss Functions & Backpropagation:** Discussion on how to measure the performance of a neural network (using loss functions like Mean Squared Error or Cross-Entropy) and the mechanism to adjust weights via backpropagation.
4. **Optimization Techniques:** Overview of algorithms to improve learning (like Gradient Descent, Stochastic Gradient Descent, and their variations) and concepts like learning rate, momentum, and regularization.
5. **Types and Applications of Deep Learning:** Introduction to various deep learning architectures like Convolutional Neural Networks (CNNs) for image processing, Recurrent Neural Networks (RNNs) for sequences, and their real-world applications.

Each of these bullet points can be expanded into more detailed sections depending on the duration and depth of the lecture.

Voice recognition and synthesis

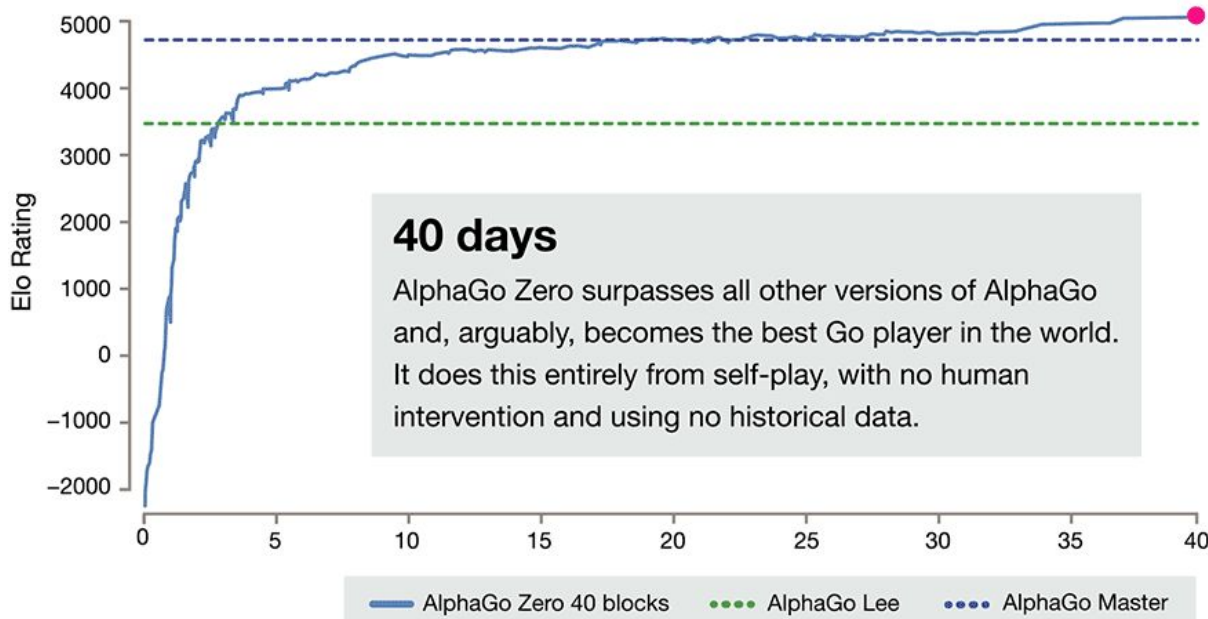
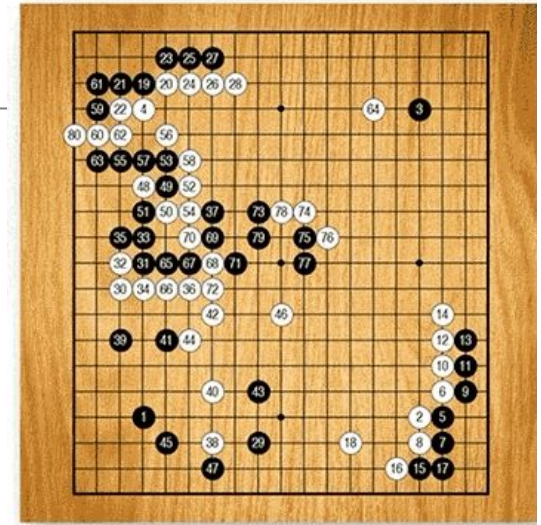


Just over a year ago we presented [WaveNet](#), a new deep neural network for generating raw audio waveforms that is capable of producing better and more realistic-sounding speech than existing techniques. At that time, the model was a research prototype and was too computationally intensive to work in consumer products.

Playing complex games

Go is a game with 10^{360} possible moves

- 2015: algorithm beating pro player considered at least 10 years away
- 2016: AlphaGo beats Lee Sedol, best Go player

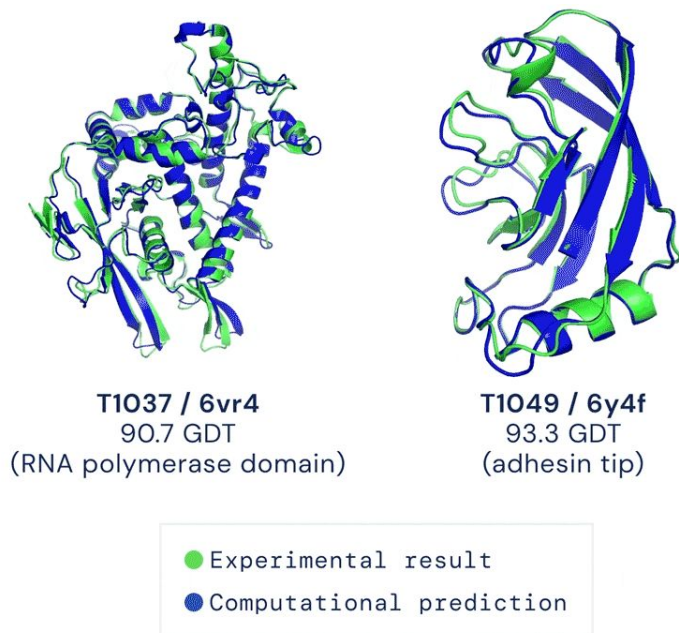


Science: Protein folding

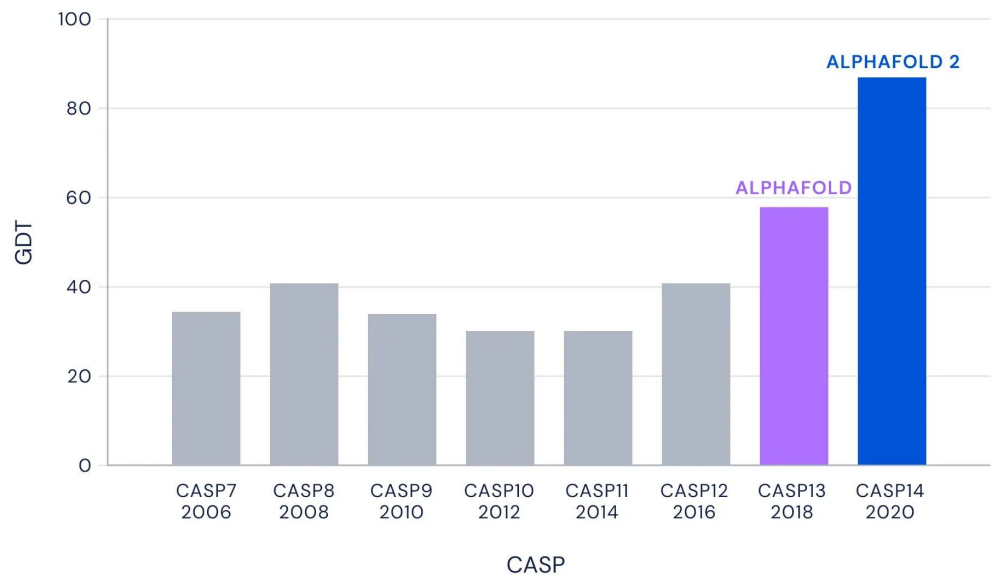
Protein folding: predict spatial configuration of protein from amino acid sequence

- Very important for structural biology, drug discovery etc.

AlphaFold2: prediction with mean error < 1 Angstrom



Median Free-Modelling Accuracy

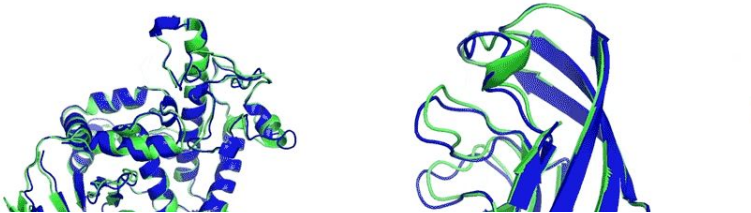


Science: Protein

Protein folding: predict spatial configuration

- Very important for structural biology

AlphaFold2: prediction with mean error



The Nobel Prize in Physics 2024

John Hopfield

“for foundational discoveries and inventions that enable machine learning with artificial neural networks”



John Hopfield, III. Niklas Elmehed © Nobel Prize Outreach

Geoffrey Hinton

“for foundational discoveries and inventions that enable machine learning with artificial neural networks”



Geoffrey Hinton, III. Niklas Elmehed © Nobel Prize Outreach

The Nobel Prize in Chemistry 2024

David Baker

“for computational protein design”



David Baker, III. Niklas Elmehed © Nobel Prize Outreach

Demis Hassabis

“for protein structure prediction”



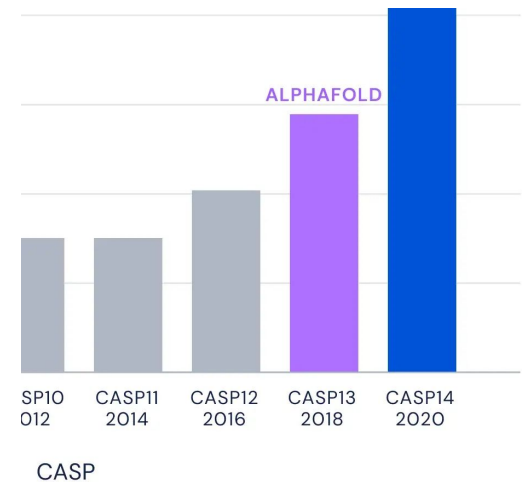
Demis Hassabis, III. Niklas Elmehed © Nobel Prize Outreach

John Jumper

“for protein structure prediction”



John Jumper, III. Niklas Elmehed © Nobel Prize Outreach

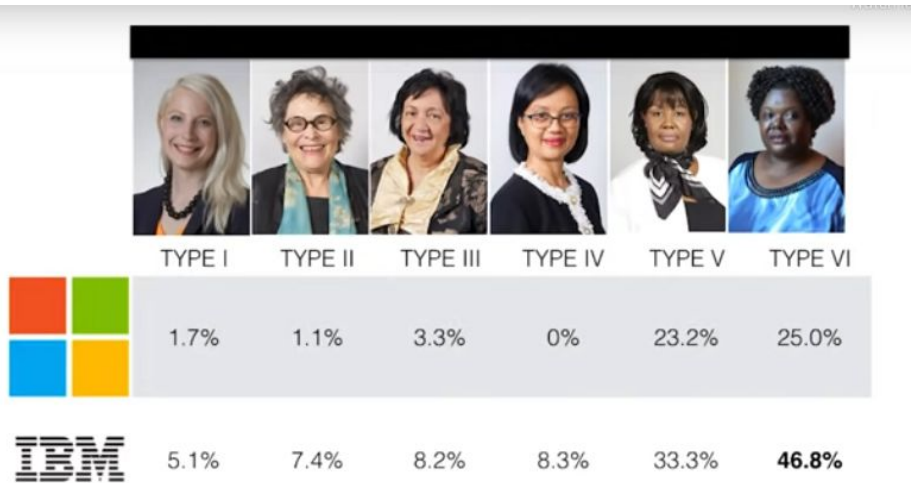


CASP

SP10 012 CASP11 2014 CASP12 2016 CASP13 2018 CASP14 2020

ALPHAFOLD

Negative Societal Impact



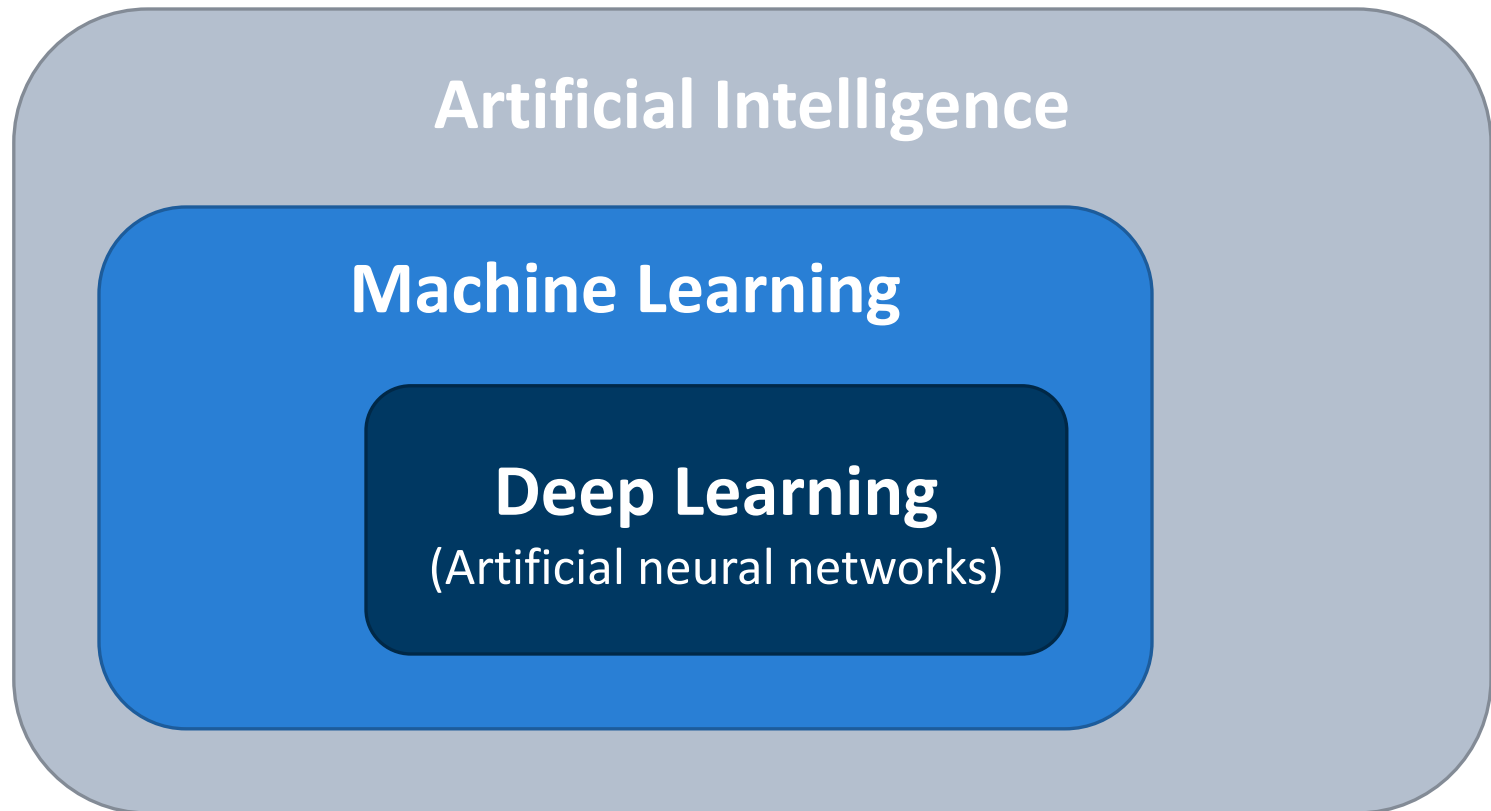
Commercial face recognition systems:
identify gender

Misclassification rate depends
significantly on ethnicity!

Training data: skewed toward
white & male



What is Deep Learning?



What is machine learning?

THE SCIENCE OF GETTING COMPUTERS TO ACT WITHOUT
BEING EXPLICITLY PROGRAMMED

Logical Reasoning vs. Machine Learning

Logical Reasoning

A $\rightarrow$ X
B $\rightarrow$ Y
C $\rightarrow$ Z

Decisions are based on predefined rules

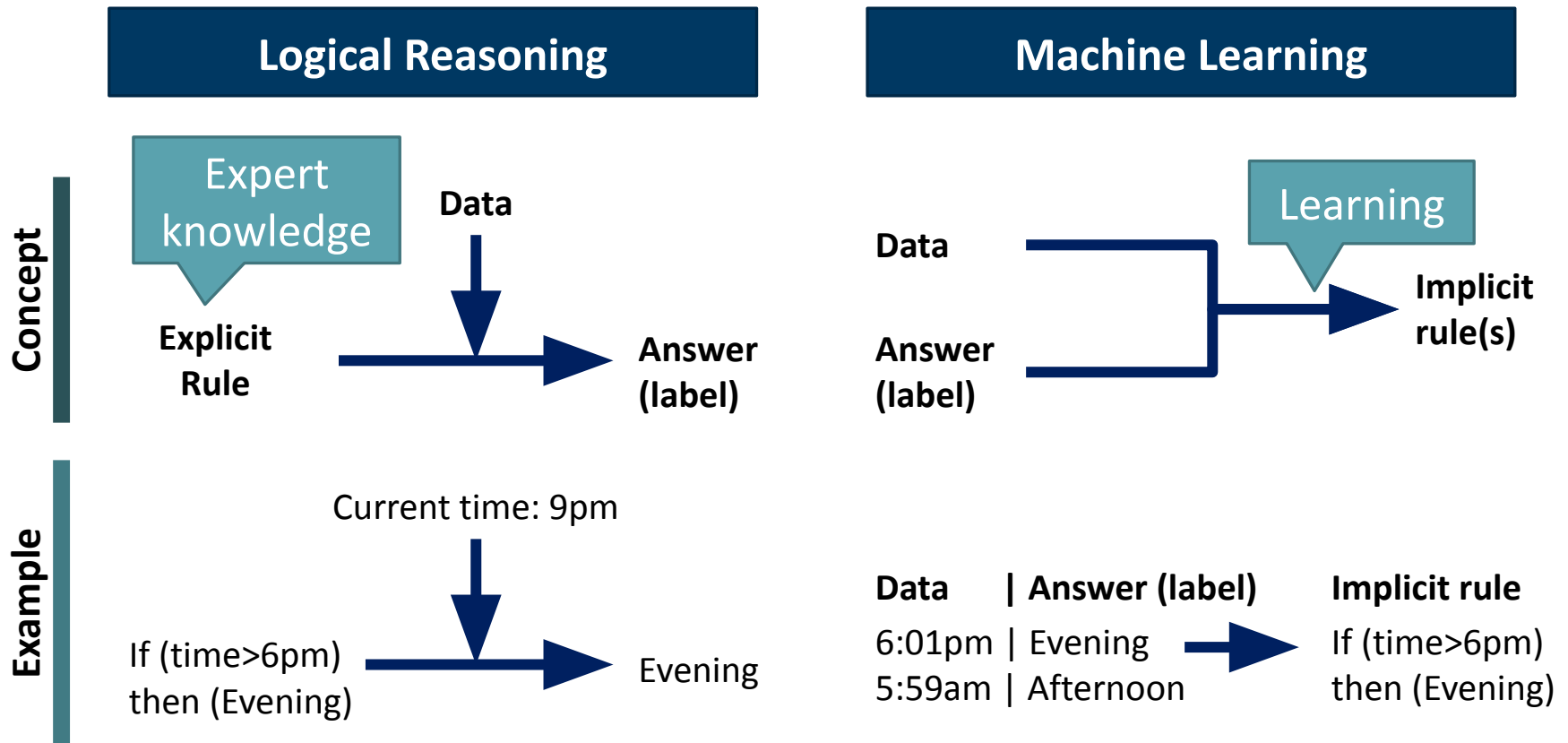
Machine Learning



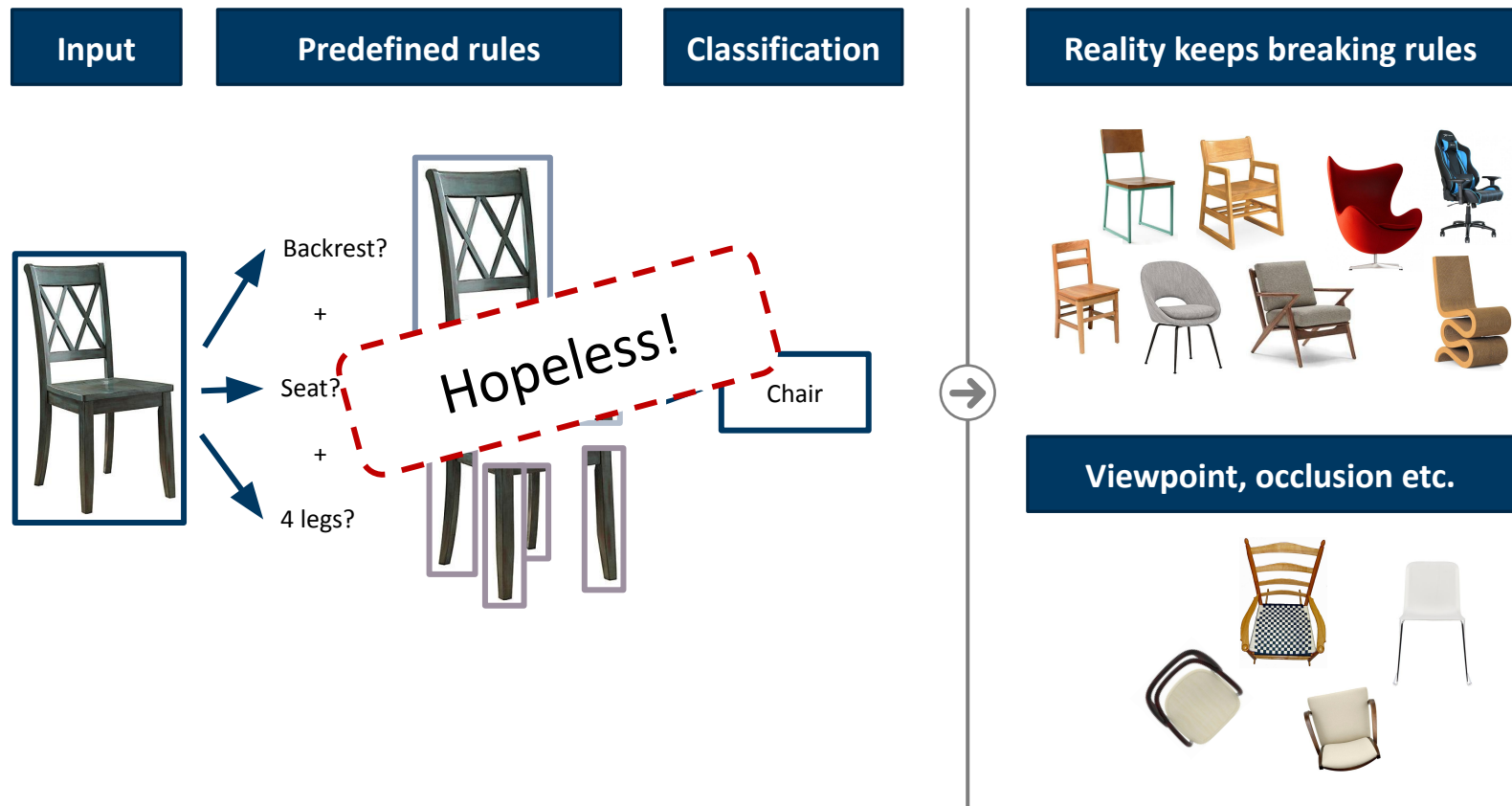
Decisions are based on a learned mapping of inputs to desired outputs

(without explicitly defining rules)

Logical Reasoning vs. Machine Learning

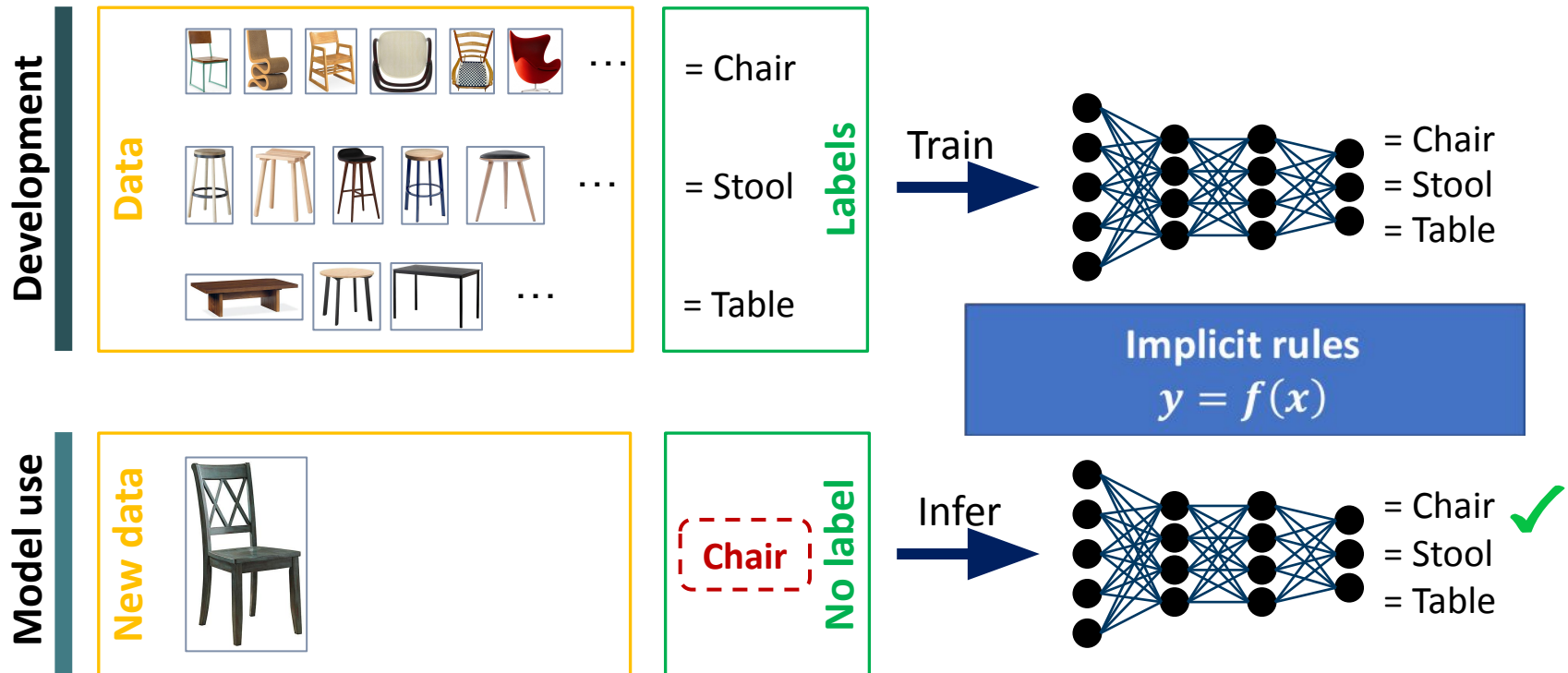


Example: object classification via classic, rule-based AI



Machine Learning:

Learn implicit rules from examples



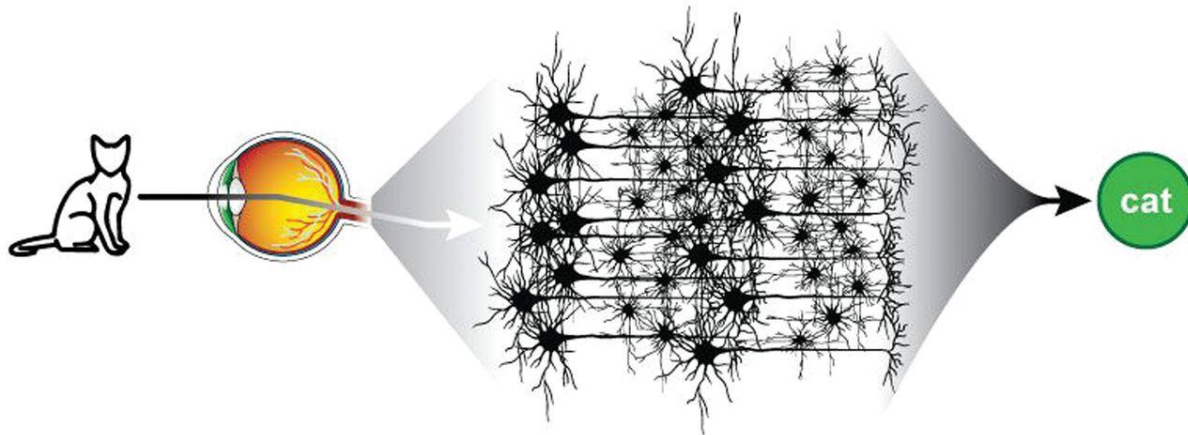
How do neural networks work?

FROM THE BRAIN TO ARTIFICIAL NEURAL NETWORKS

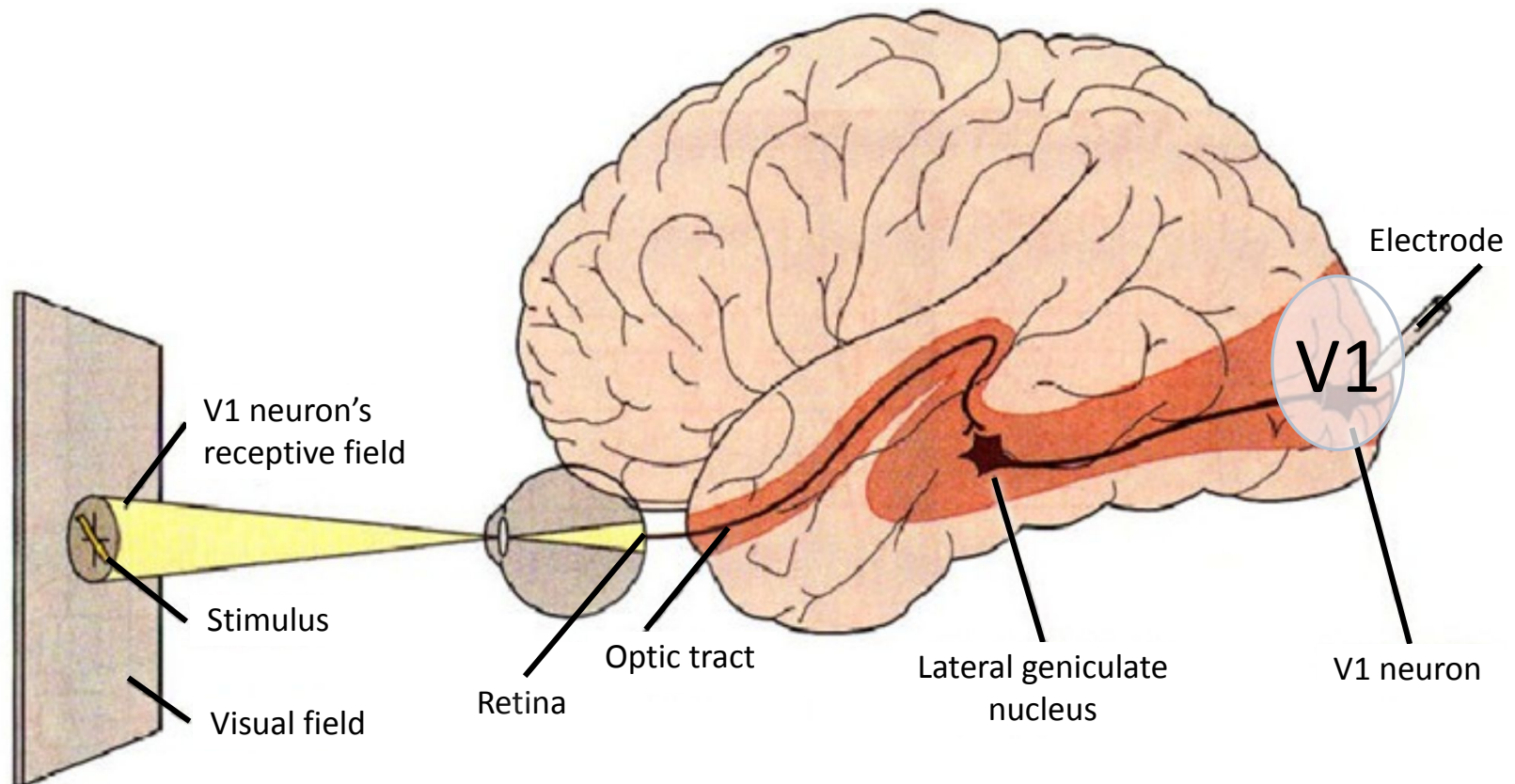
Biological neural networks

Information processing in the brain:
networks of neurons that communicate through electrical impulses

Example: vision

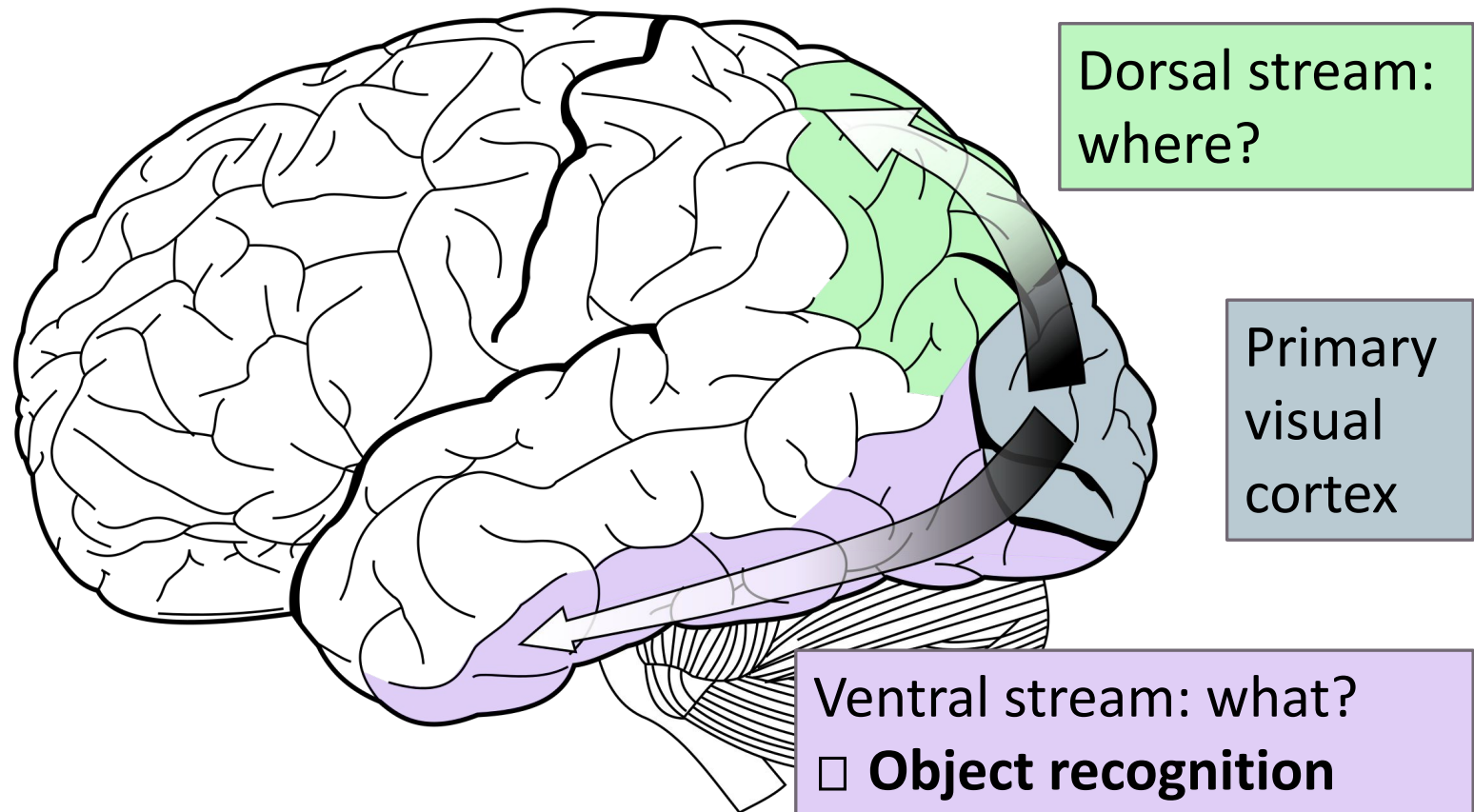


From the eye to the brain

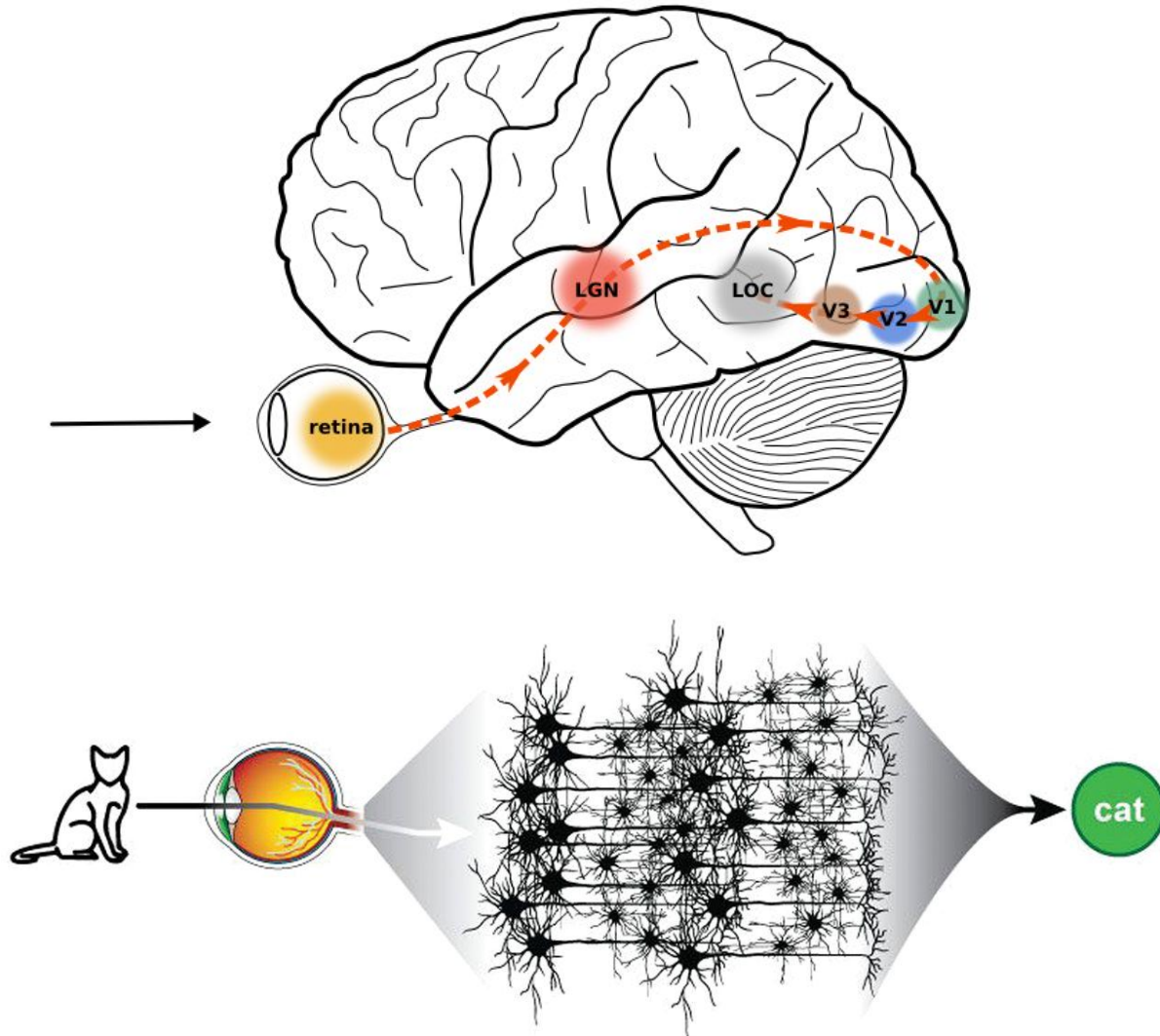


V1 = Primary visual cortex

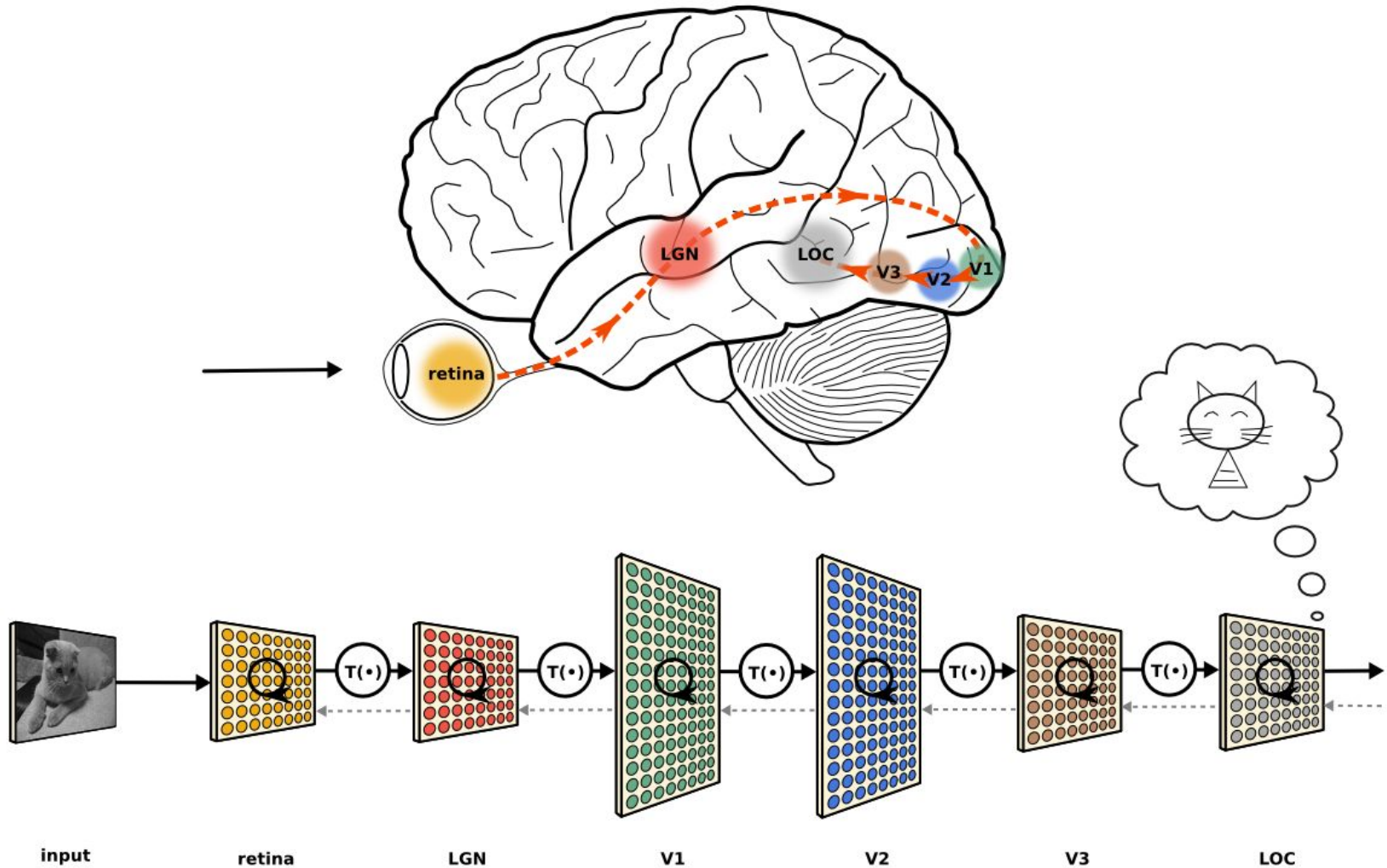
The visual system: two streams



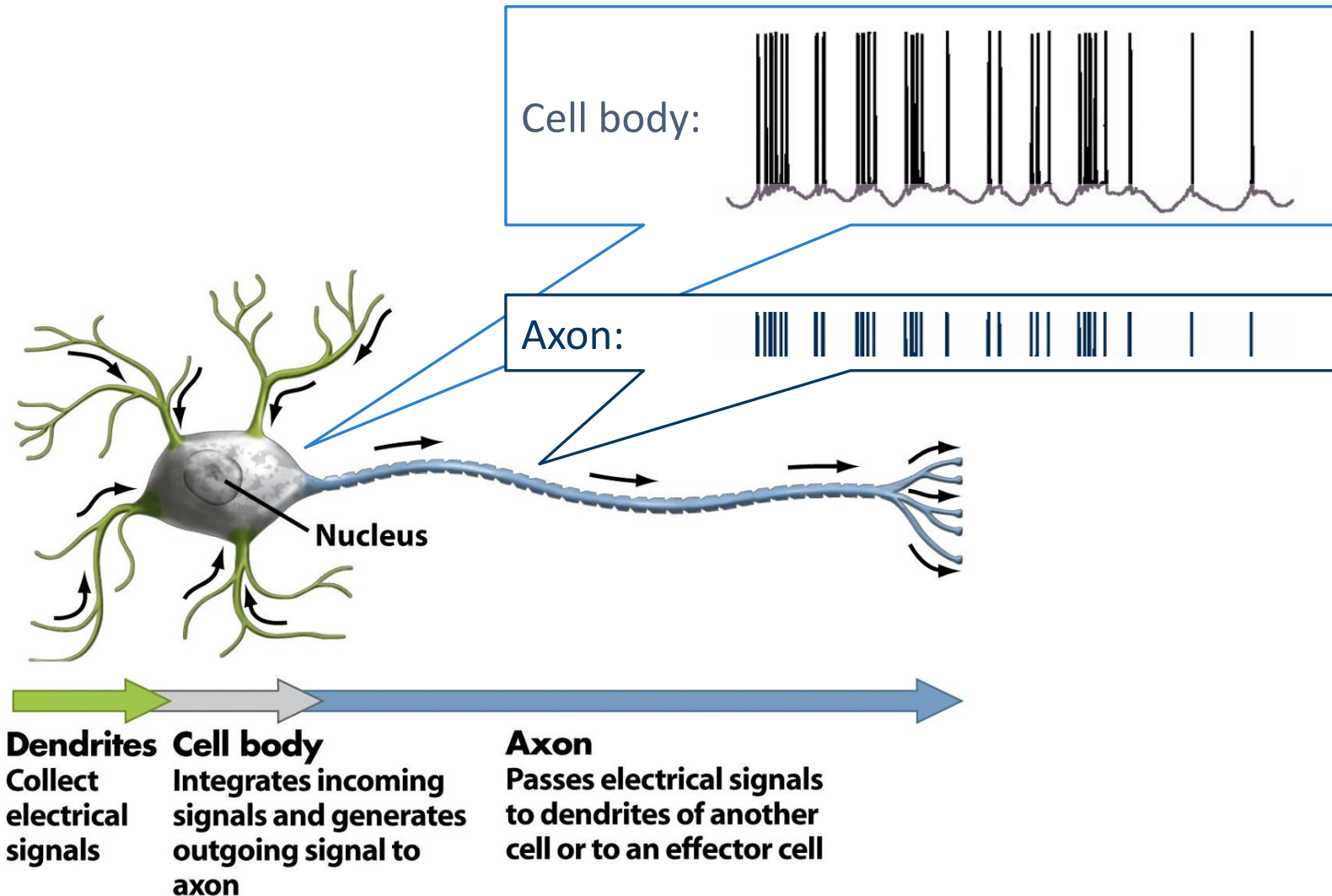
Hierarchical data processing



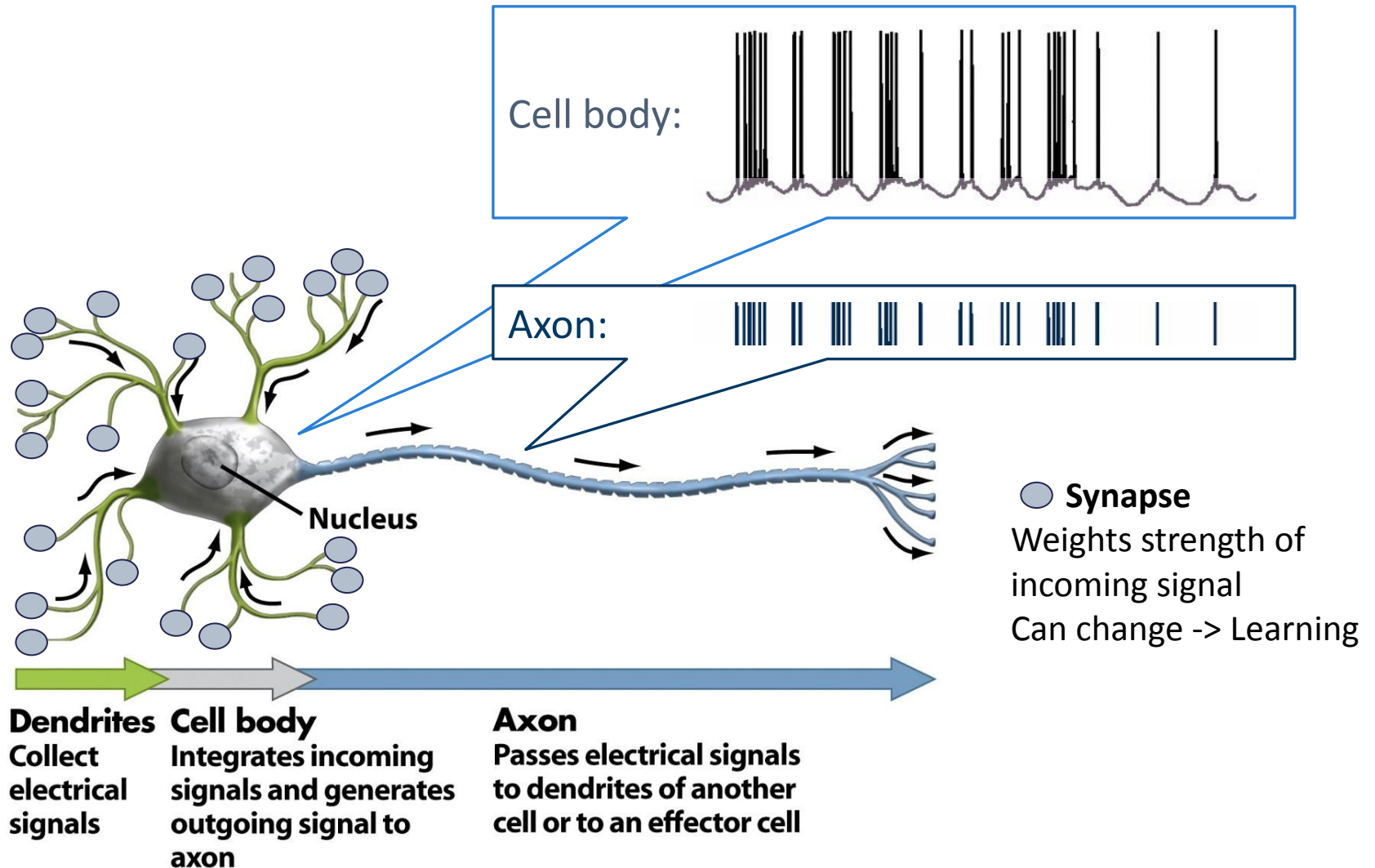
Hierarchical data processing



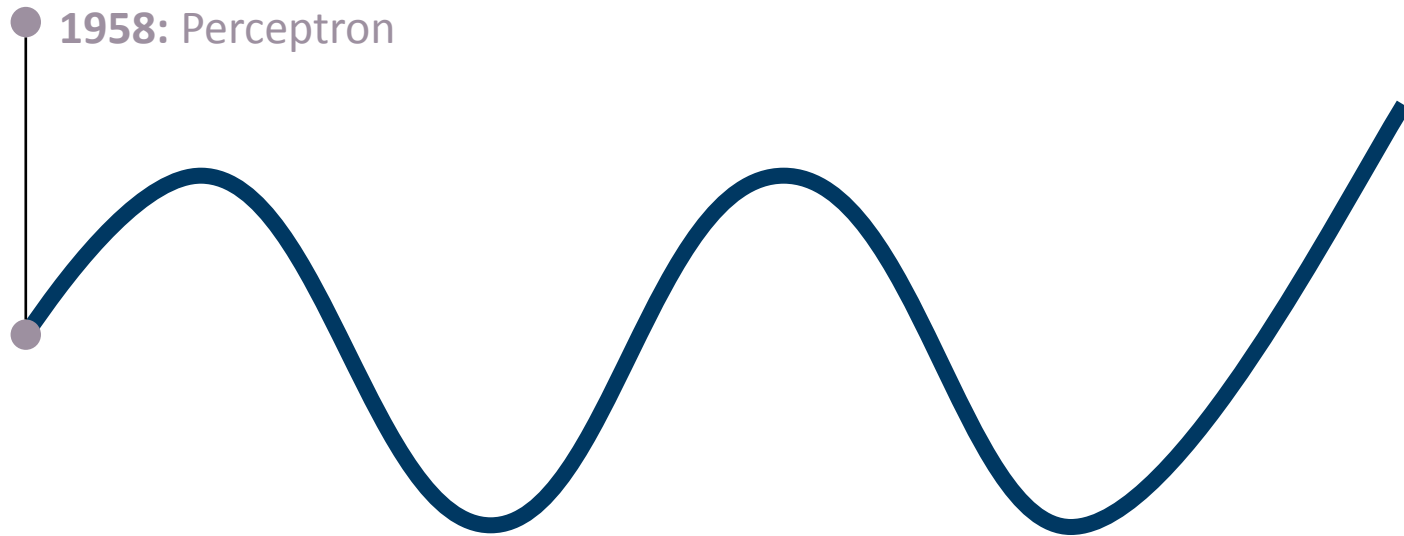
How do neurons communicate?



How do neurons communicate?



Neural networks: summers & winters



The Perceptron (Rosenblatt 1958)

NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo
of Computer Designed to
Read and Grow Wiser

WASHINGTON, July 7 (UPI)—The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

The embryo—the Weather Bureau's \$2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's demonstration for newsmen.

The service said it would use this principle to build the first of its Perceptron thinking machines that will be able to read and write. It is expected to be finished in about a year at a cost of \$100,000.

Dr. Frank Rosenblatt, designer of the Perceptron, conducted the demonstration. He said the machine would be the first device to think as the human brain. As do human be-

ings, Perceptron will make mistakes at first, but will grow wiser as it gains experience, he said.

Dr. Rosenblatt, a research psychologist at the Cornell Aeronautical Laboratory, Buffalo, said Perceptrons might be fired to the planets as mechanical space explorers.

Without Human Controls

The Navy said the perceptron would be the first non-living mechanism "capable of receiving, recognizing and identifying its surroundings without any human training or control."

The "brain" is designed to remember images and information it has perceived itself. Ordinary computers remember only what is fed into them on punch cards or magnetic tape.

Later Perceptrons will be able to recognize people and call out their names and instantly translate speech in one language to speech or writing in another language, it was predicted.

Mr. Rosenblatt said in principle it would be possible to build brains that could reproduce themselves on an assembly line and which would be conscious of their existence.

The New York Times

July 7, 1958

In today's demonstration, the "704" was fed two cards, one with squares marked on the left side and the other with squares on the right side.

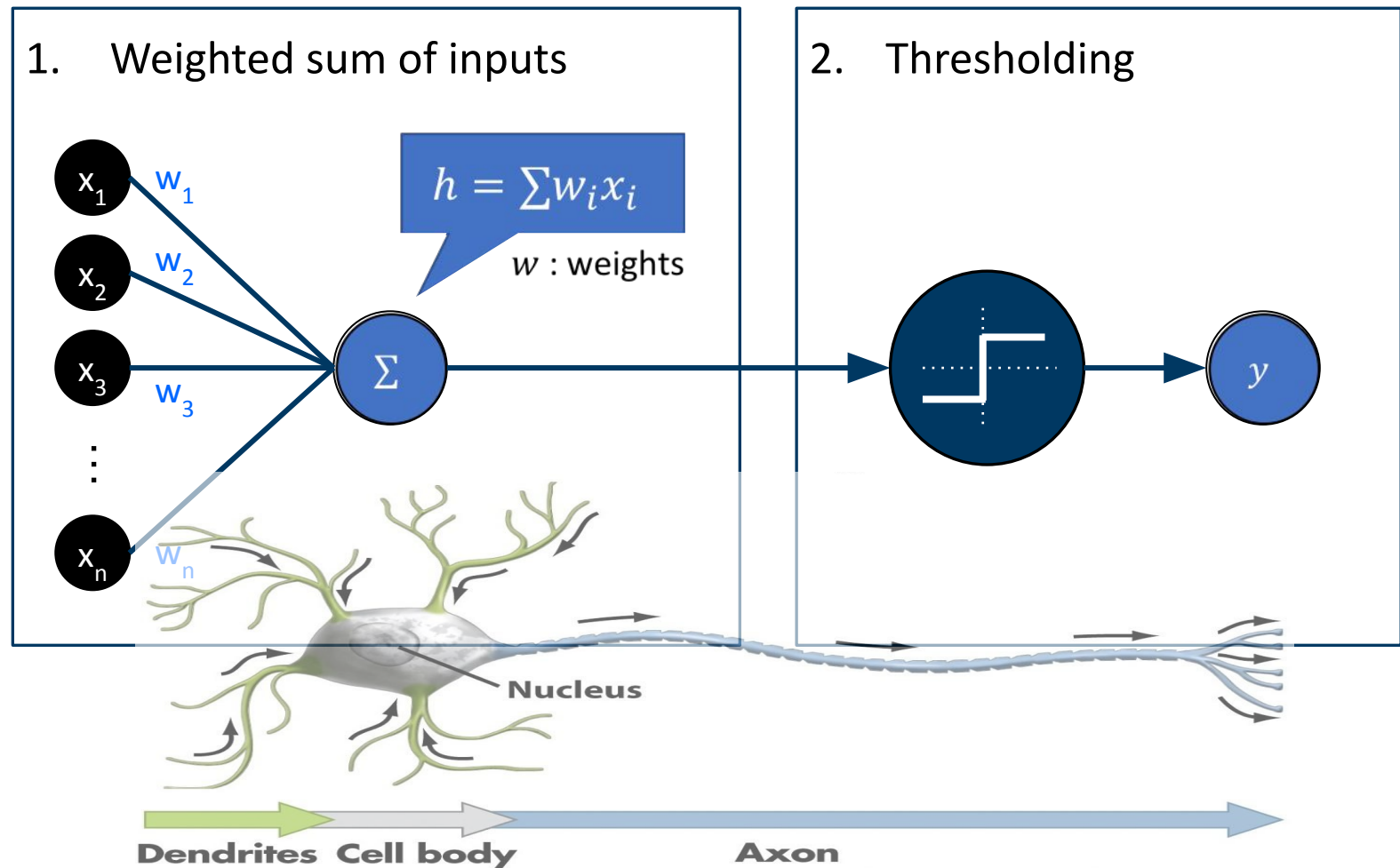
Learns by Doing

In the first fifty trials, the machine made no distinction between them. It then started registering a "Q" for the left squares and "O" for the right squares.

Dr. Rosenblatt said he could explain why the machine learned only in highly technical terms. But he said the computer had undergone a "self-induced change in the wiring diagram."

The first Perceptron will have about 1,000 electronic "association cells" receiving electrical impulses from an eye-like scanning device with 400 photo-cells. The human brain has 10,000,000,000 responsive cells, including 100,000,000 connections with the eyes.

The Perceptron (Rosenblatt 1958)

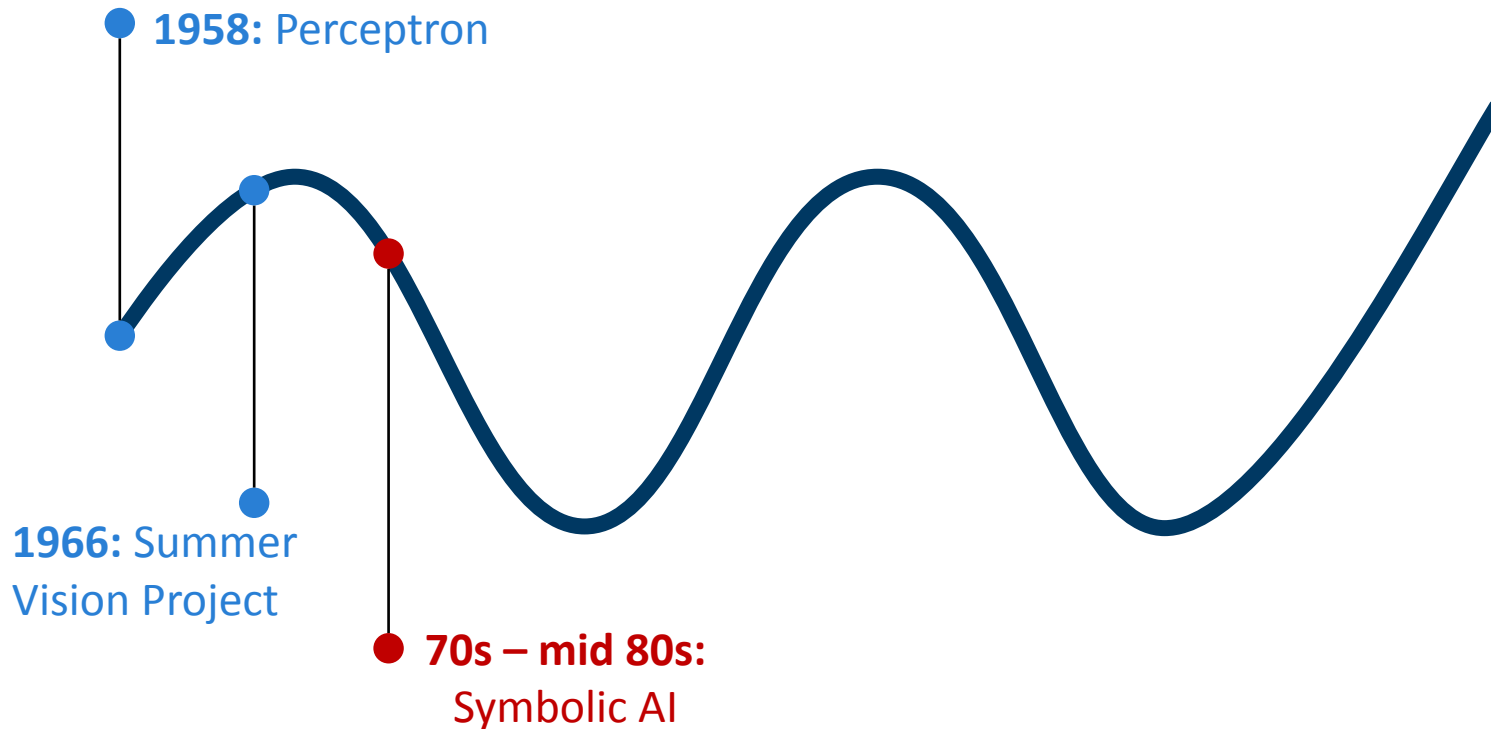


The Summer Vision Project (Papert & Minsky 1966)

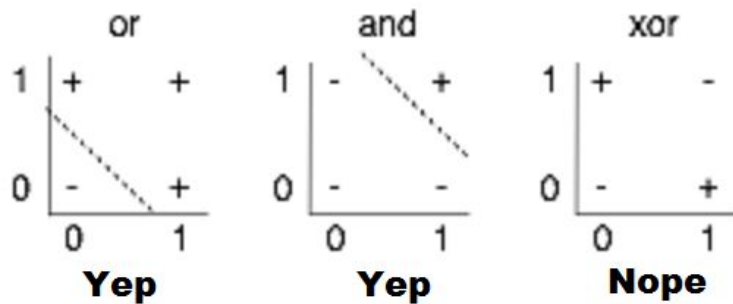
“The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of pattern recognition.”

S. Papert 1966

Neural networks: summers & winters

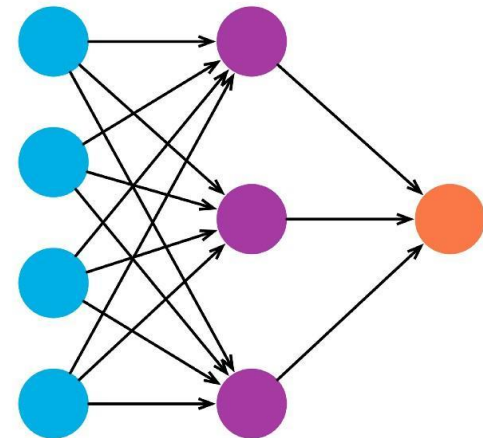


The first AI winter: 1969–1986



Perceptron

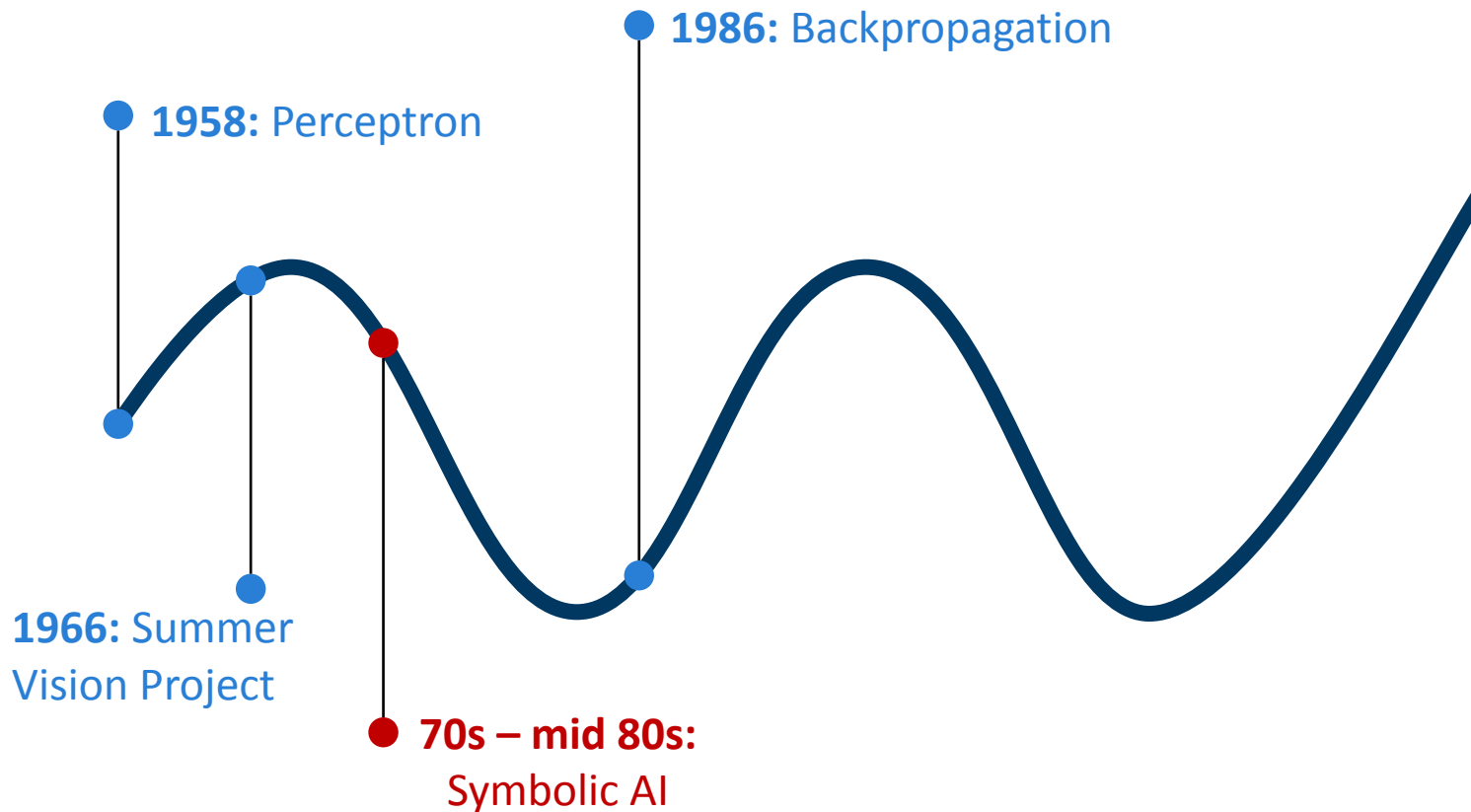
Cannot solve XOR



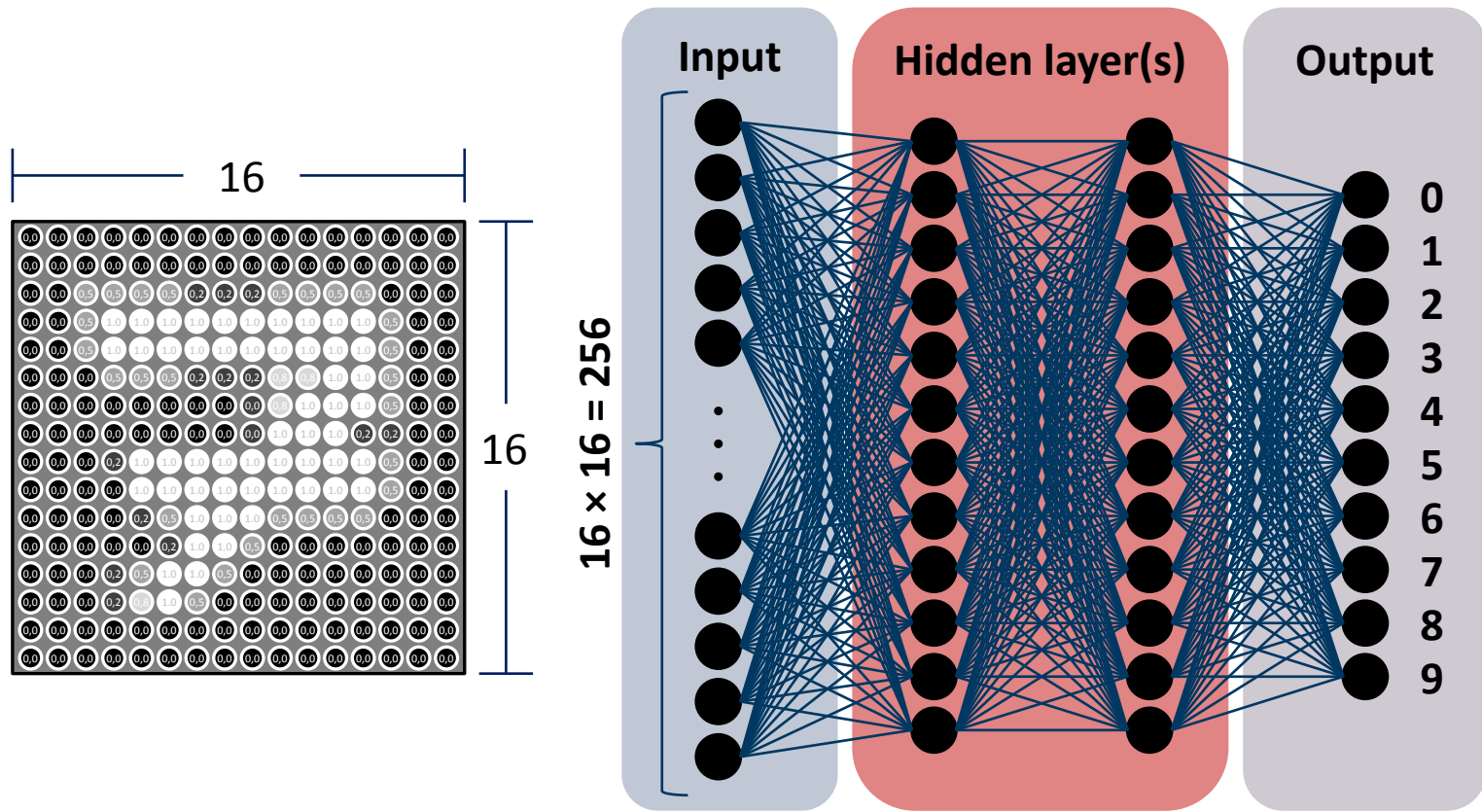
Multi-layer neural net

Problem: how to train?

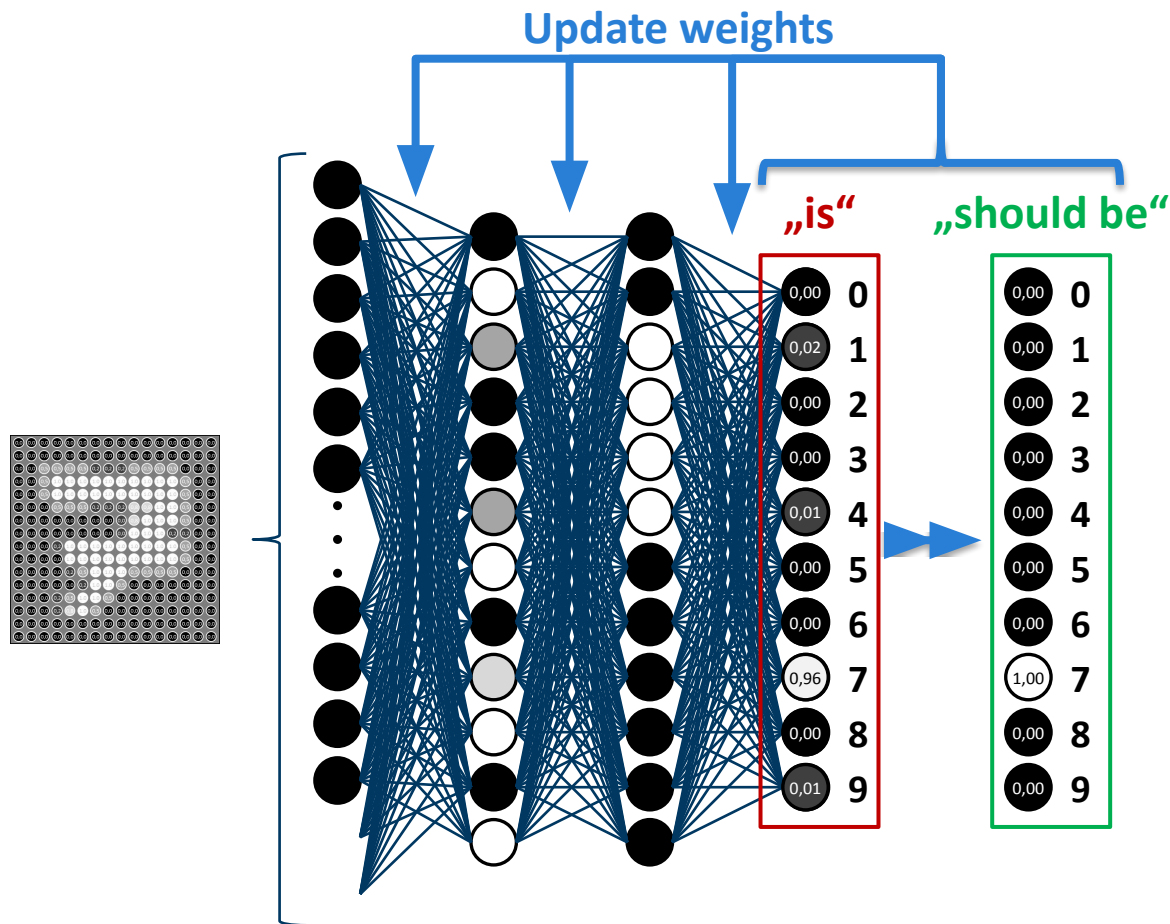
Neural networks: summers & winters



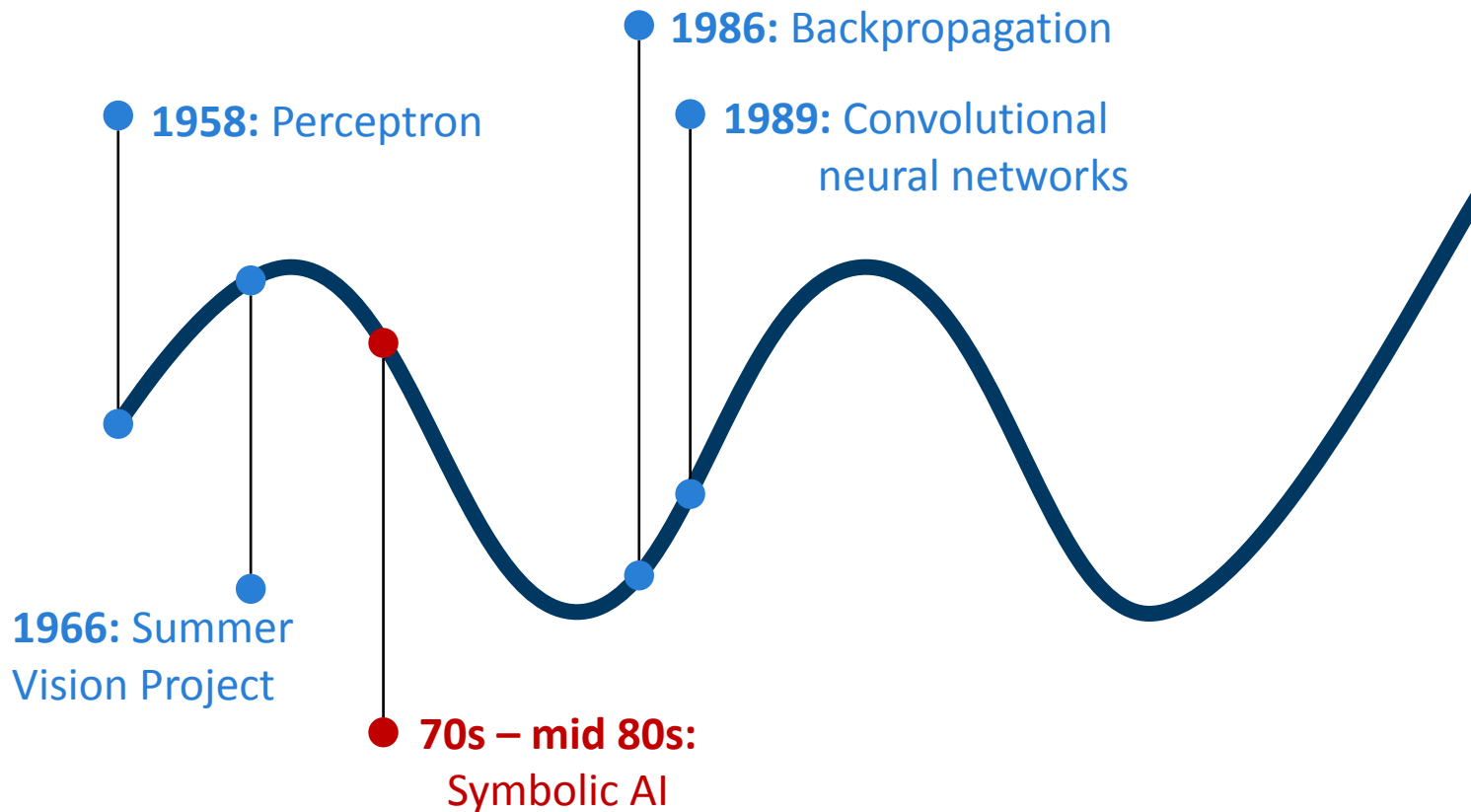
Multi-layer neural networks



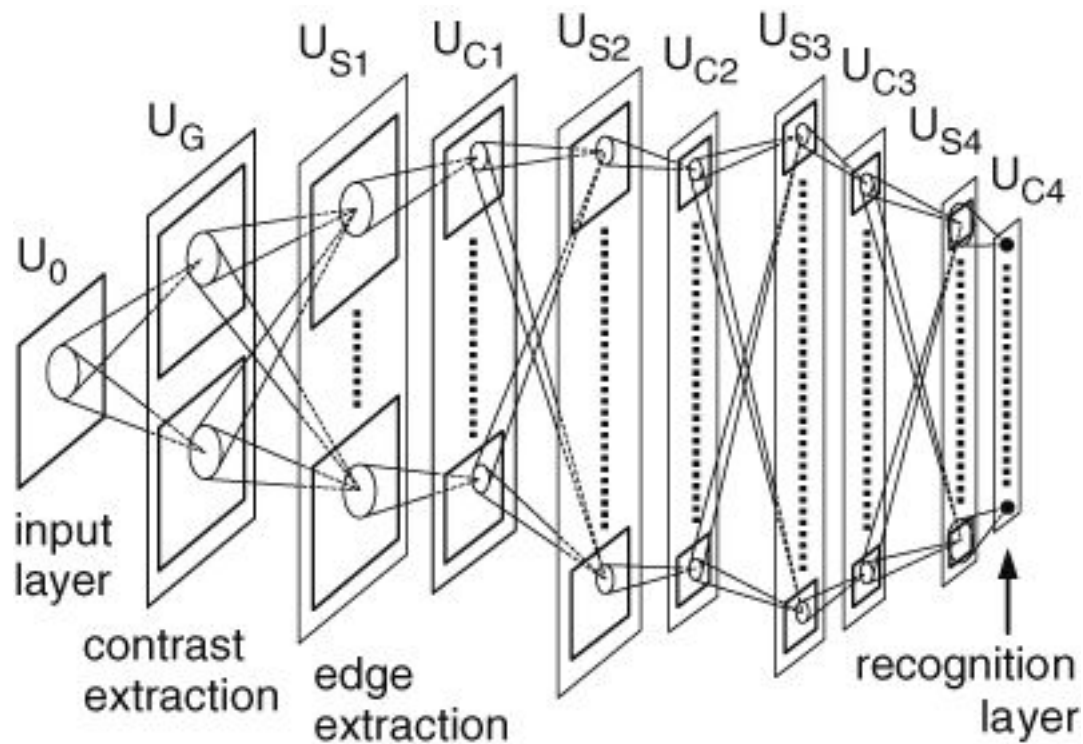
Training through backpropagation



Neural networks: summers & winters

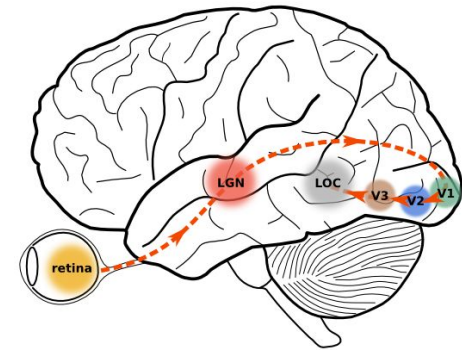
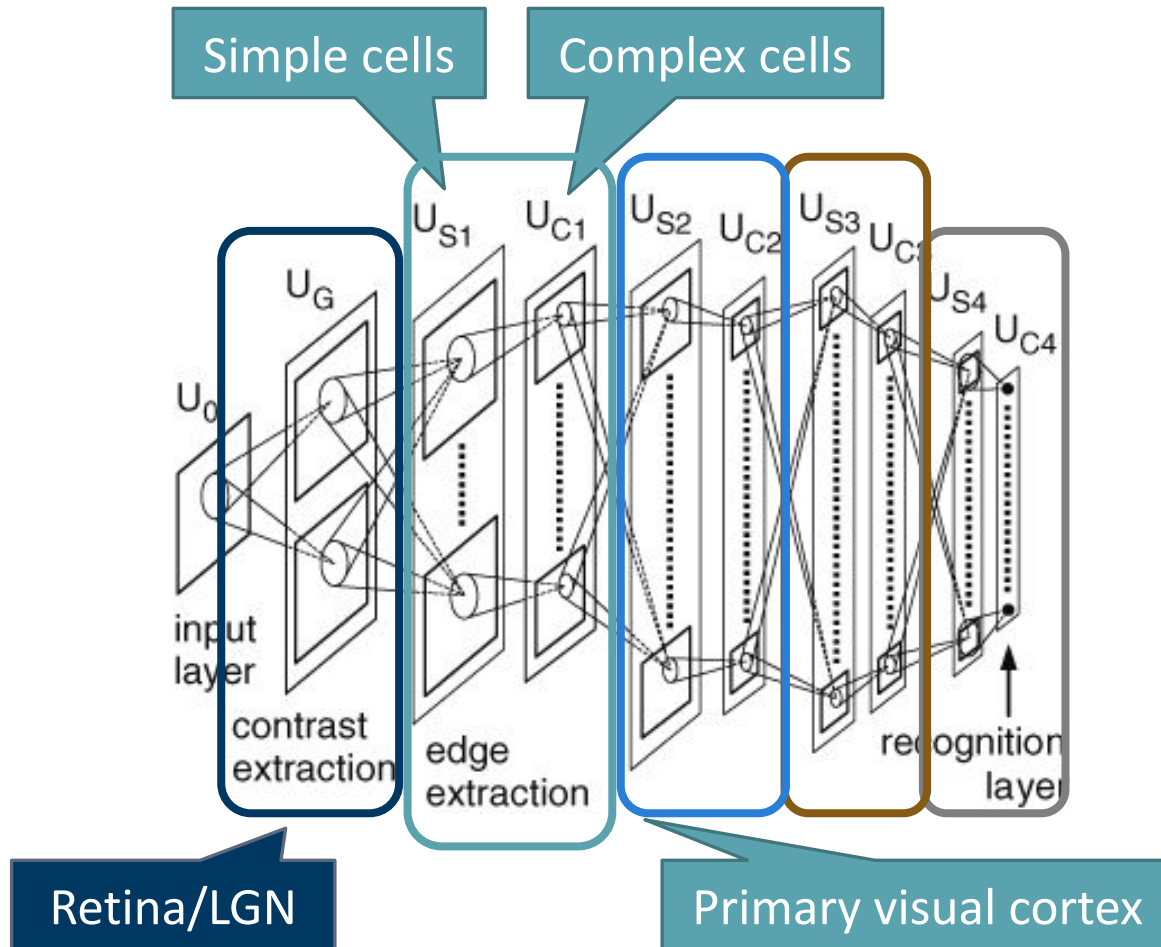


The Neocognitron (Fukushima 1980)

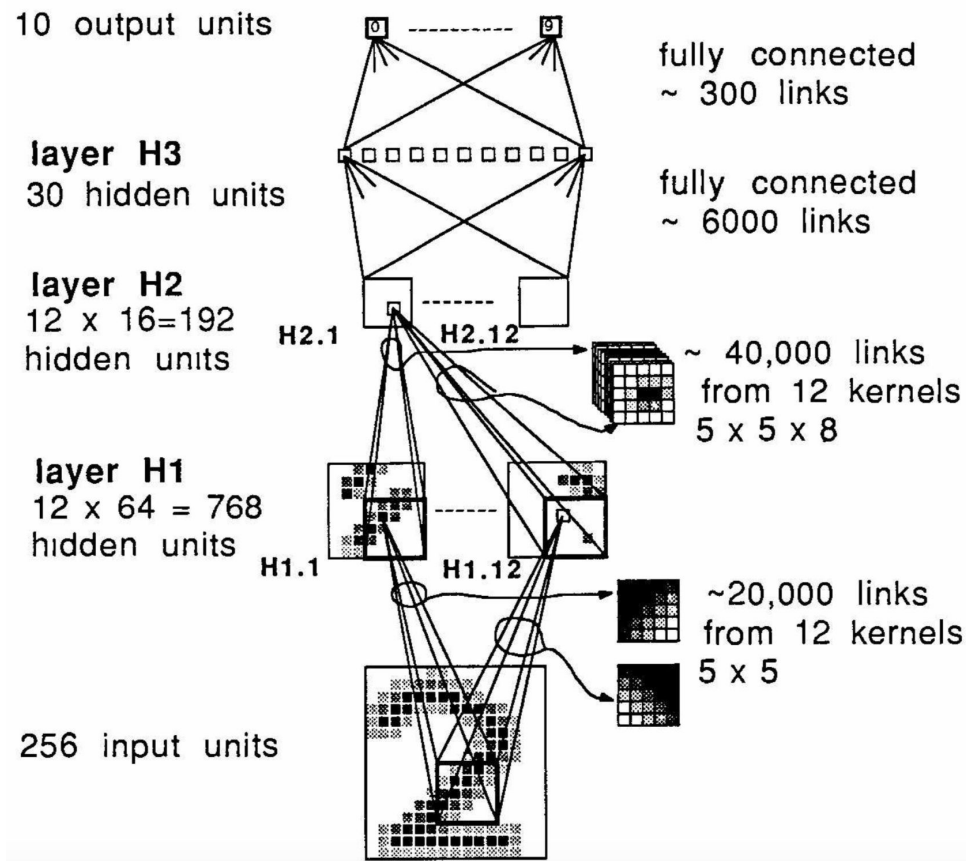


K. Fukushima (1980): "Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position", *Biological Cybernetics*.

The Neocognitron (Fukushima 1980)

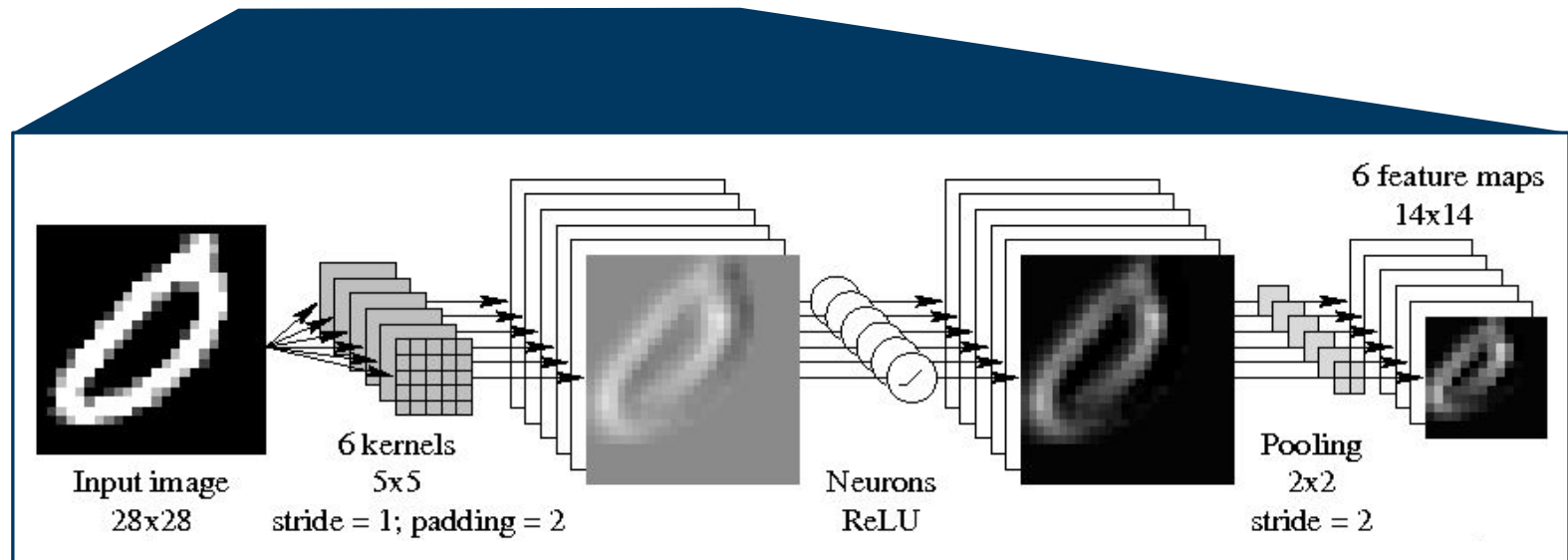
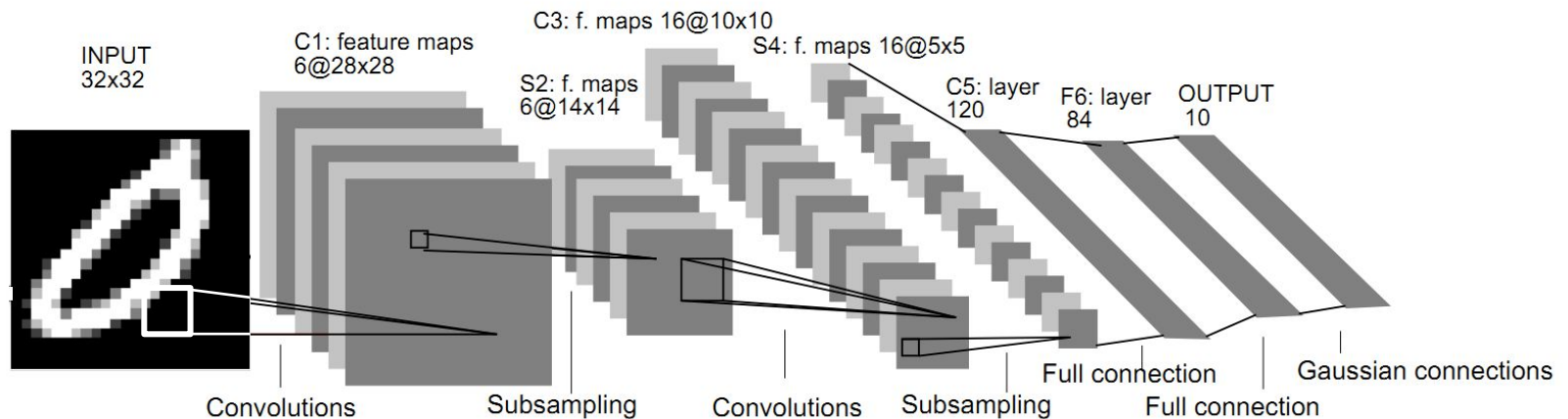


Convolutional neural networks

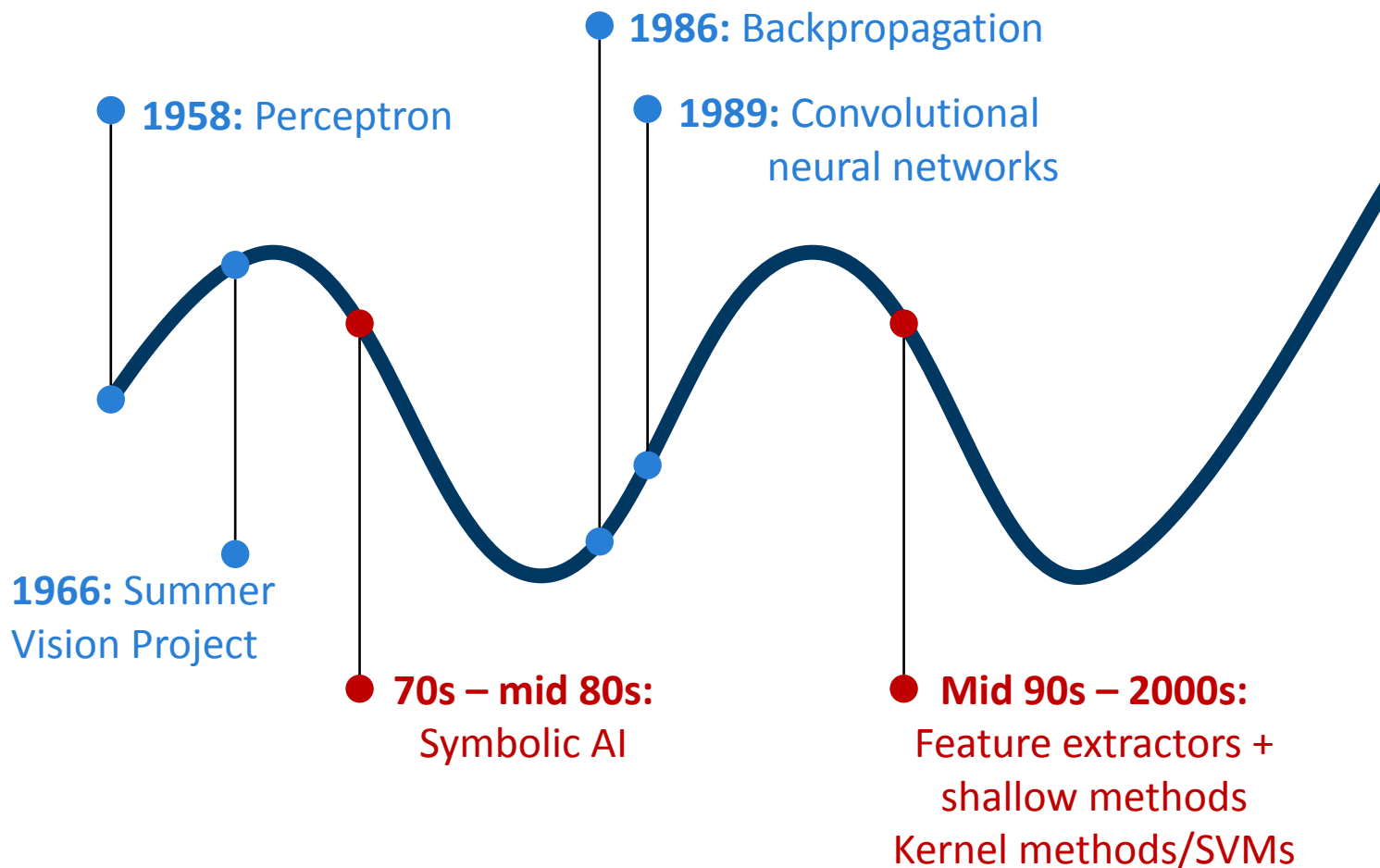


Used in production
for reading checks
and addresses

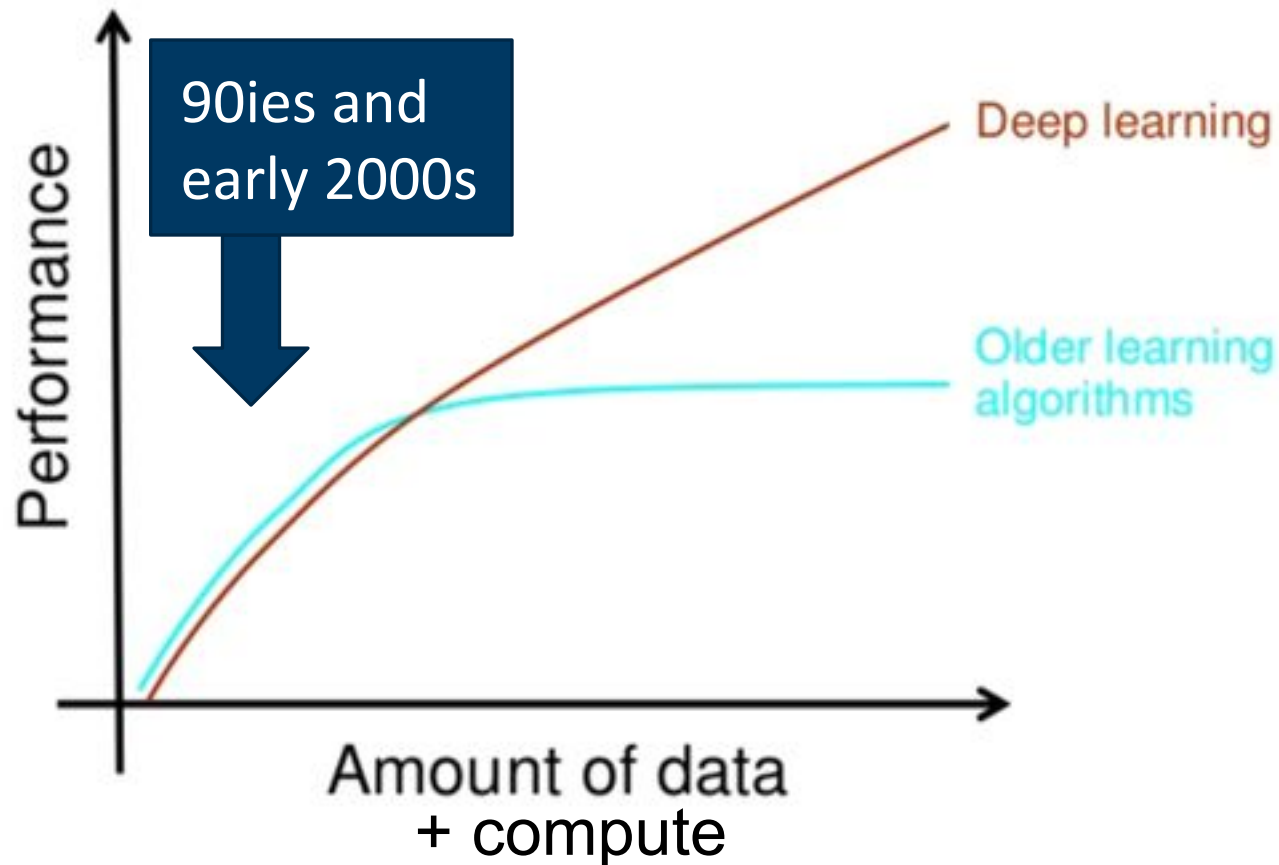
Convolutional neural networks



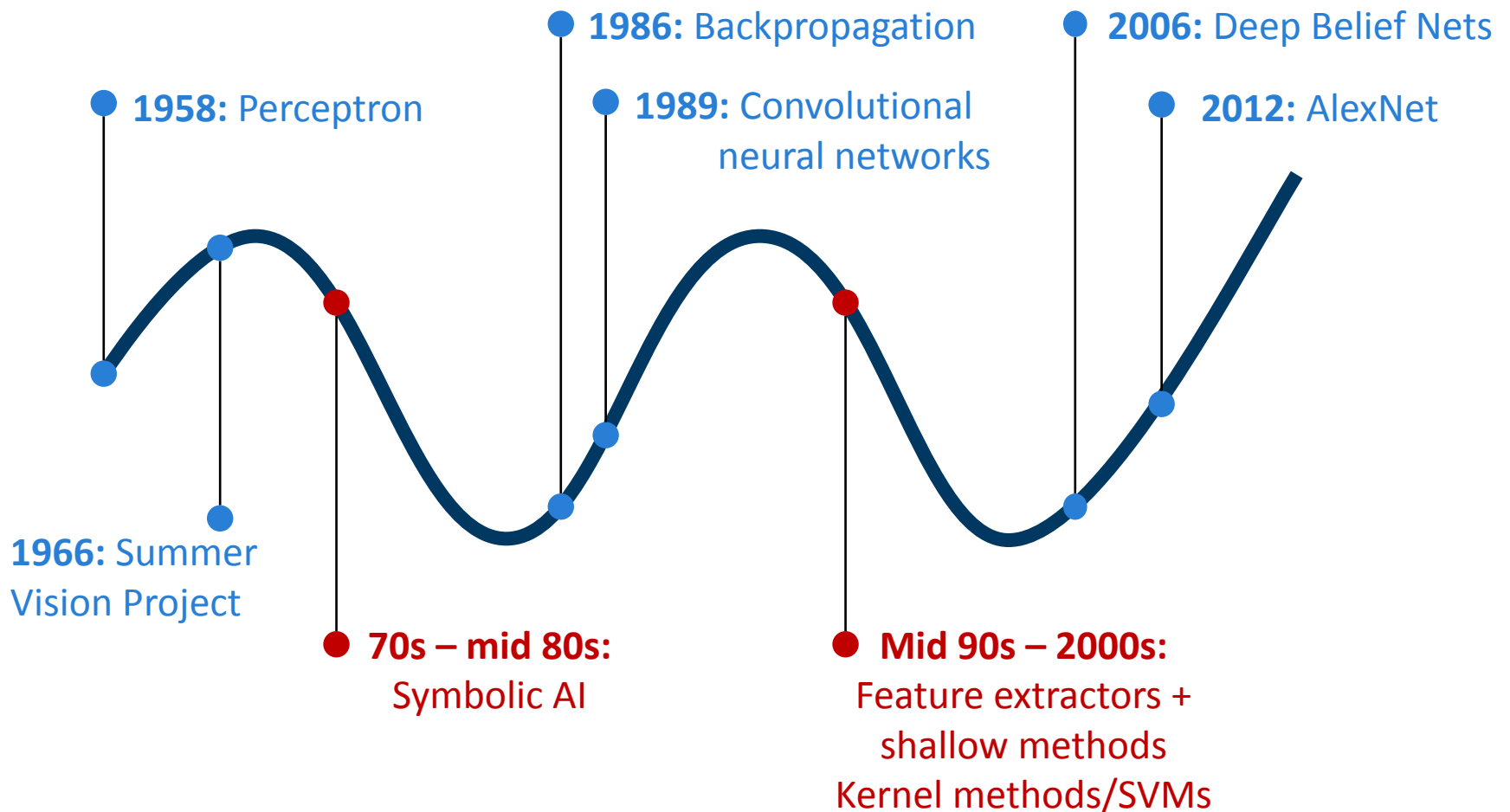
Another neural network winter



90ies: Too little data & compute power for neural networks to shine



Neural networks: summers & winters



Object recognition: IMAGENET

Dataset and benchmark on image classification

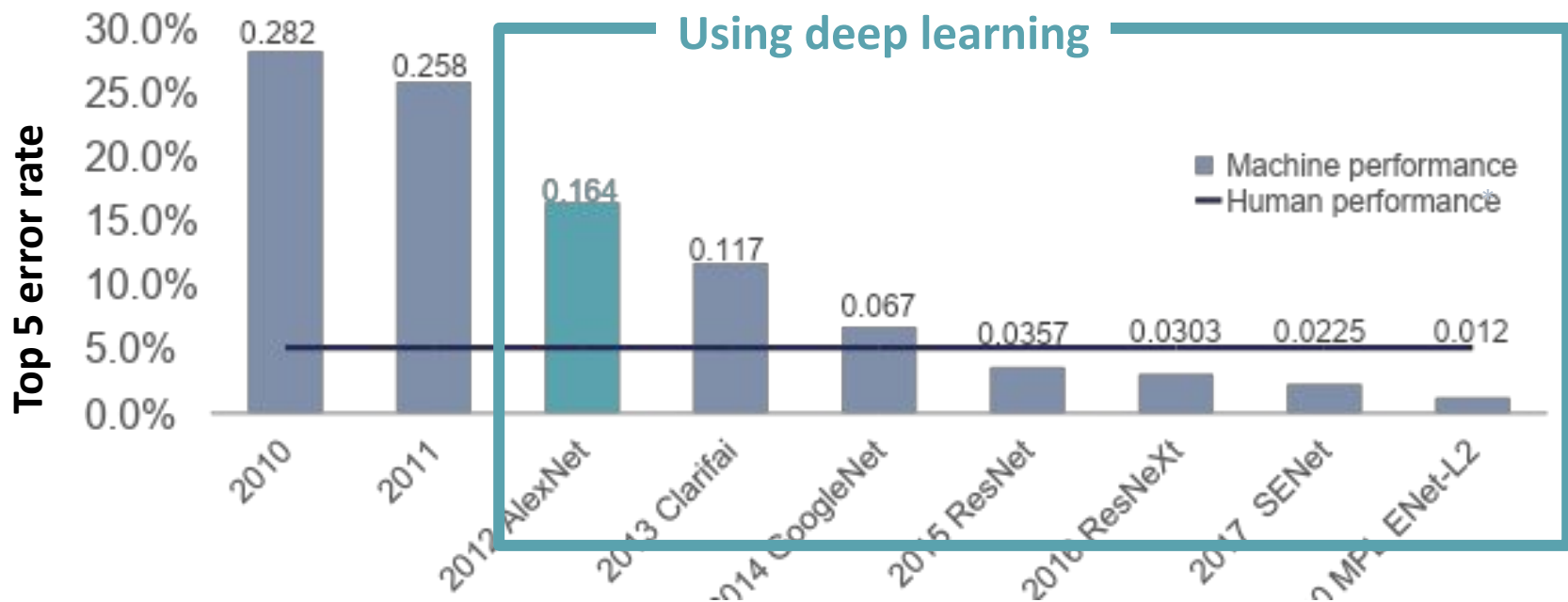
- 1 million images with ground truth class labels for training (hand-annotated)
- 1000 object categories

Deng et al., CVPR 2009



The breakthrough: “AlexNet” 2012

IMAGENET



Does this mean vision is solved?

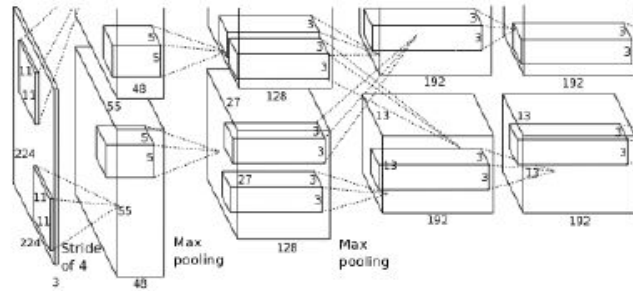
So why does it work now?

The Deep Learning Recipe for Computer Vision



Big Data: ImageNet

+



Deep Convolutional Neural Network

+

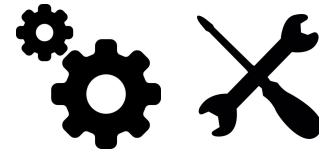


Backprop on GPU

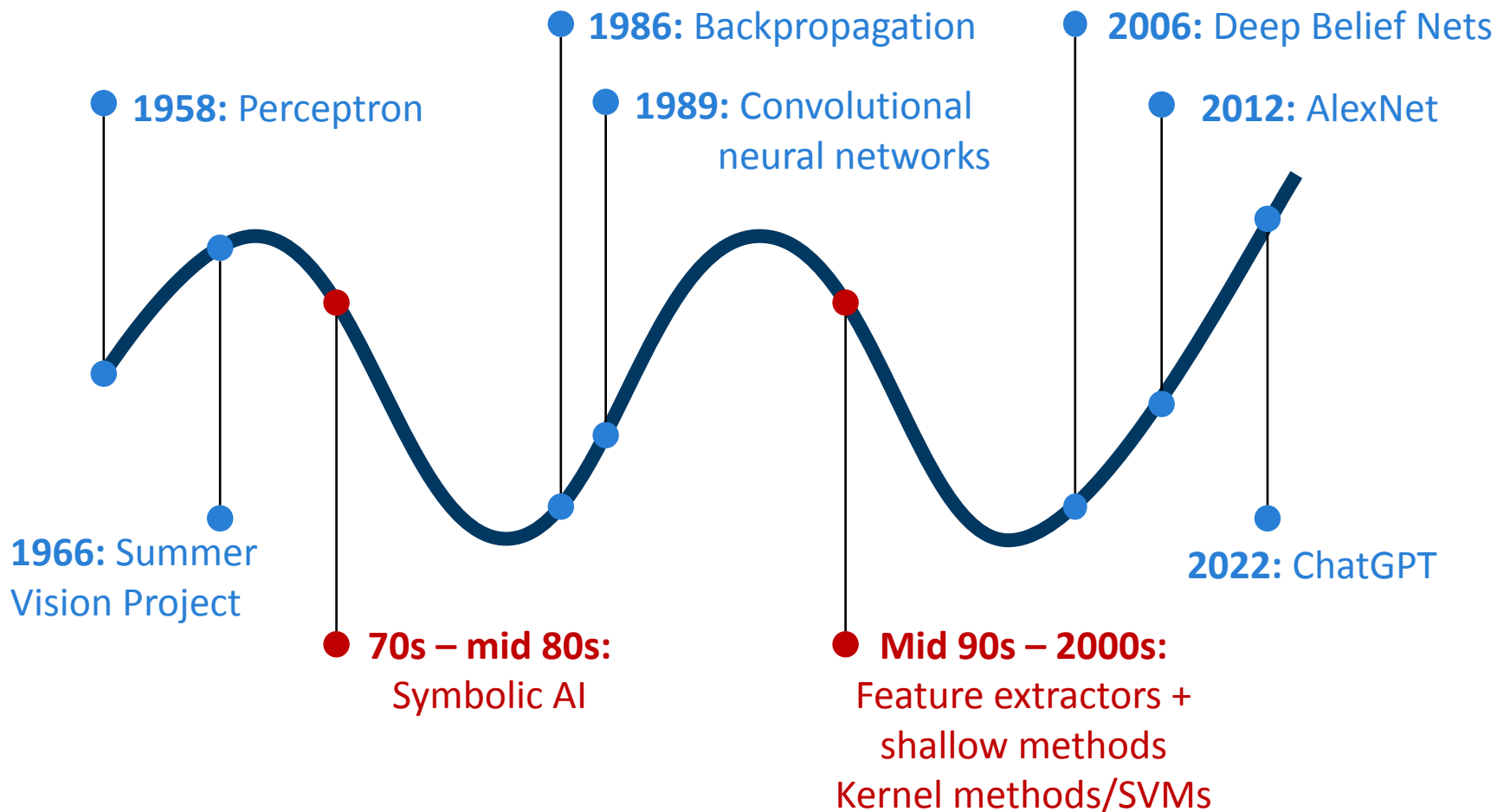
+

Many small tweaks

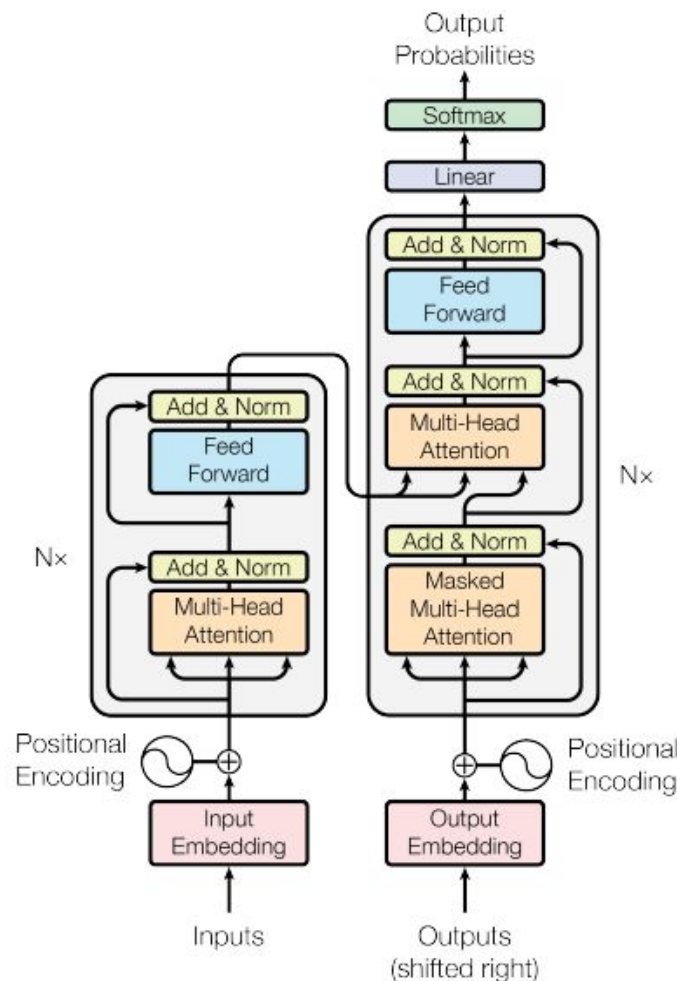
Sigmoid ☐ ReLU, batch normalization, dropout regularization



Neural networks: summers & winters



Transformers, LLMs, ChatGPT



Foundation / Frontier Models

- Large (Transformer) Models trained on massive datasets
 - significant financial investment (Google, OpenAI, Meta, ...)
- Language: trained to predict next part of the text
- Text & Vision: multimodal models

Deep learning = brain?

NO.

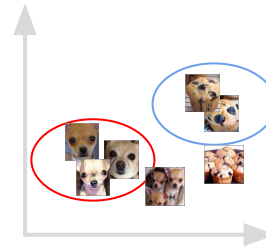
Learning paradigms

Supervised Learning



Dog
or
Muffin?

Unsupervised Learning



Reinforcement Learning



Description

Learn mapping from input to labeled output

Model underlying structure of dataset

Learn strategy to get reward when interacting with the environment

Feedback

Correct answer is given (labeled data)

No answer is given (unlabeled data)

Occasional reward, but no direct feedback on individual actions

Example

- Image classification
- Machine translation

- Clustering
- Dimensionality reduction

- Learning to play chess
- Robot learning to perform a task

Learning paradigms

Most significant advances with deep learning

Supervised Learning



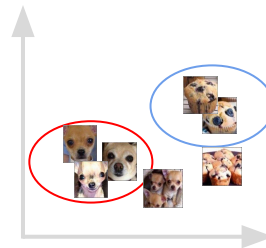
Dog or Muffin?

Learn mapping from input to labeled output

Correct answer is given (labeled data)

- Image classification
- Machine translation

Unsupervised Learning



Model underlying structure of dataset

No answer is given (unlabeled data)

- Clustering
- Dimensionality reduction

Reinforcement Learning



Learn strategy to get reward when interacting with the environment

Occasional reward, but no direct feedback on individual actions

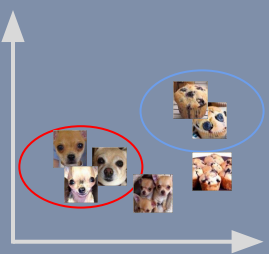
- Learning to play chess
- Robot learning to perform a task

Description

Feedback

Example

Learning paradigms

	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Description	 Advances in recent years enable foundation models (self-supervised learning) Learn mapping from input to labeled output	 Model underlying structure of dataset	 Learn strategy to get reward when interacting with the environment
Feedback	Correct answer is given (labeled data)	No answer is given (unlabeled data)	Occasional reward, but no direct feedback on individual actions
Example	• Image classification • Machine translation	• Clustering • Dimensionality reduction	• Learning to play chess • Robot learning to perform a task

Is vision solved? “Human-level” performance on ImageNet

ImageNet: 1 million images, 1,000 classes

Top 5-error rate: “average” human ~5%, Best CNN: <1%

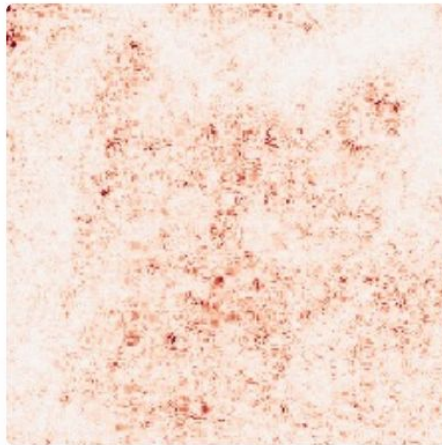


Differences between human and machine: „adversarial examples“



“Tiger cat”

+



Adversarial perturbation
(contrast 10x exaggerated)

=



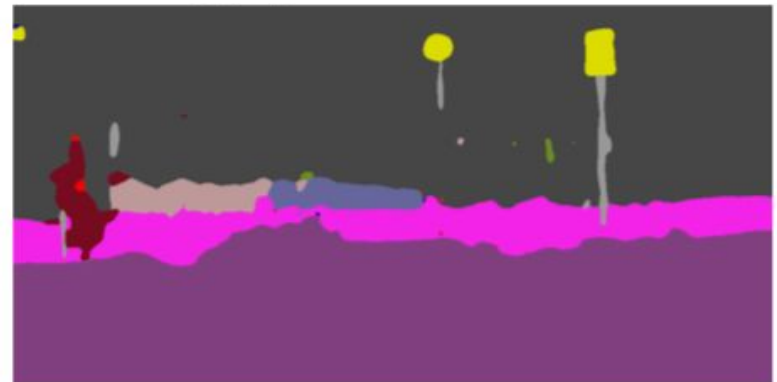
“Ostrich”

Self-driving cars?

(a) original image



(b) adv. example



Real-world example



Turtle or rifle?

Differences part II: cat or elephant?



(a) Texture image

81.4%	Indian elephant
10.3%	indri
8.2%	black swan



(b) Content image

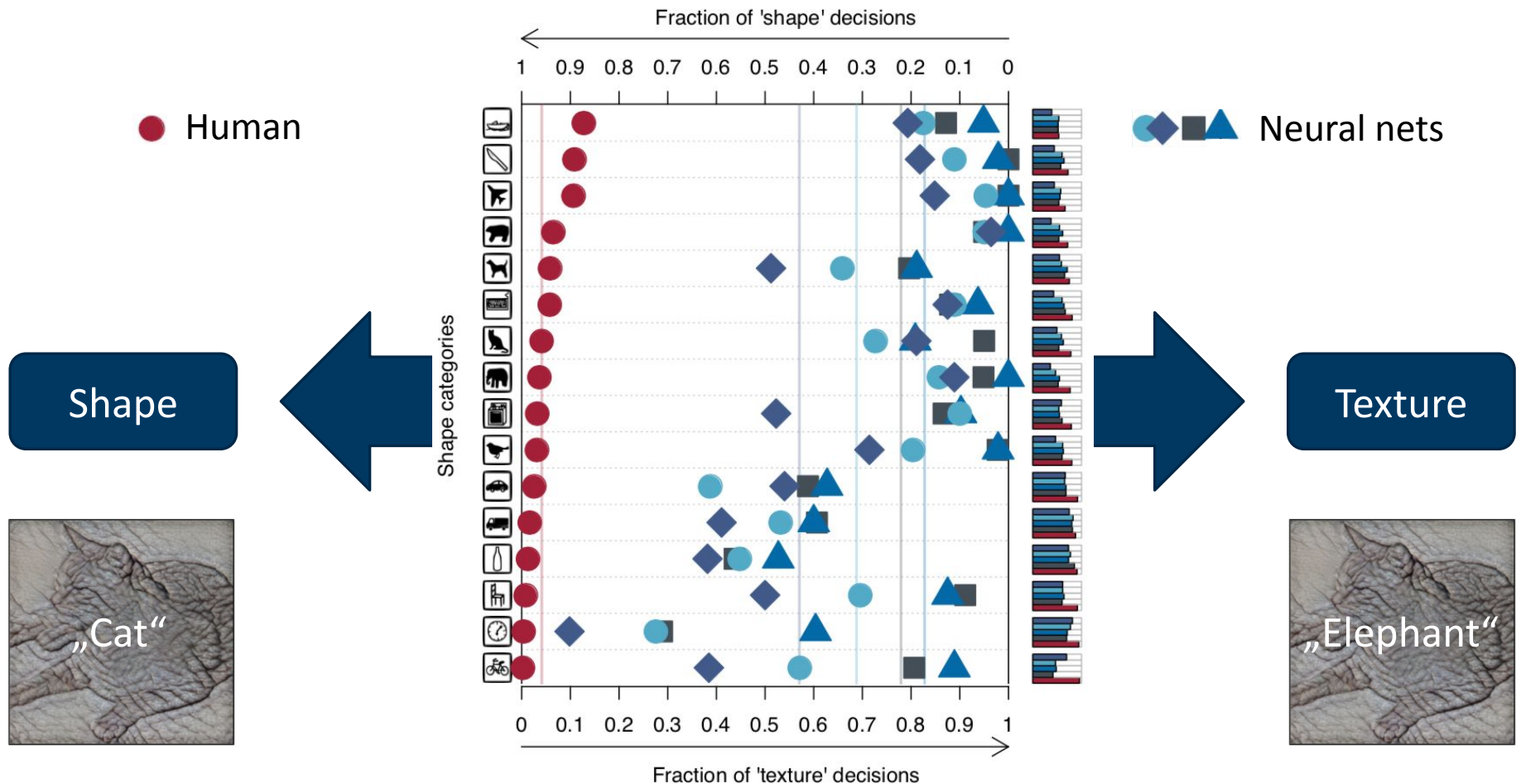
71.1%	tabby cat
17.3%	grey fox
3.3%	Siamese cat



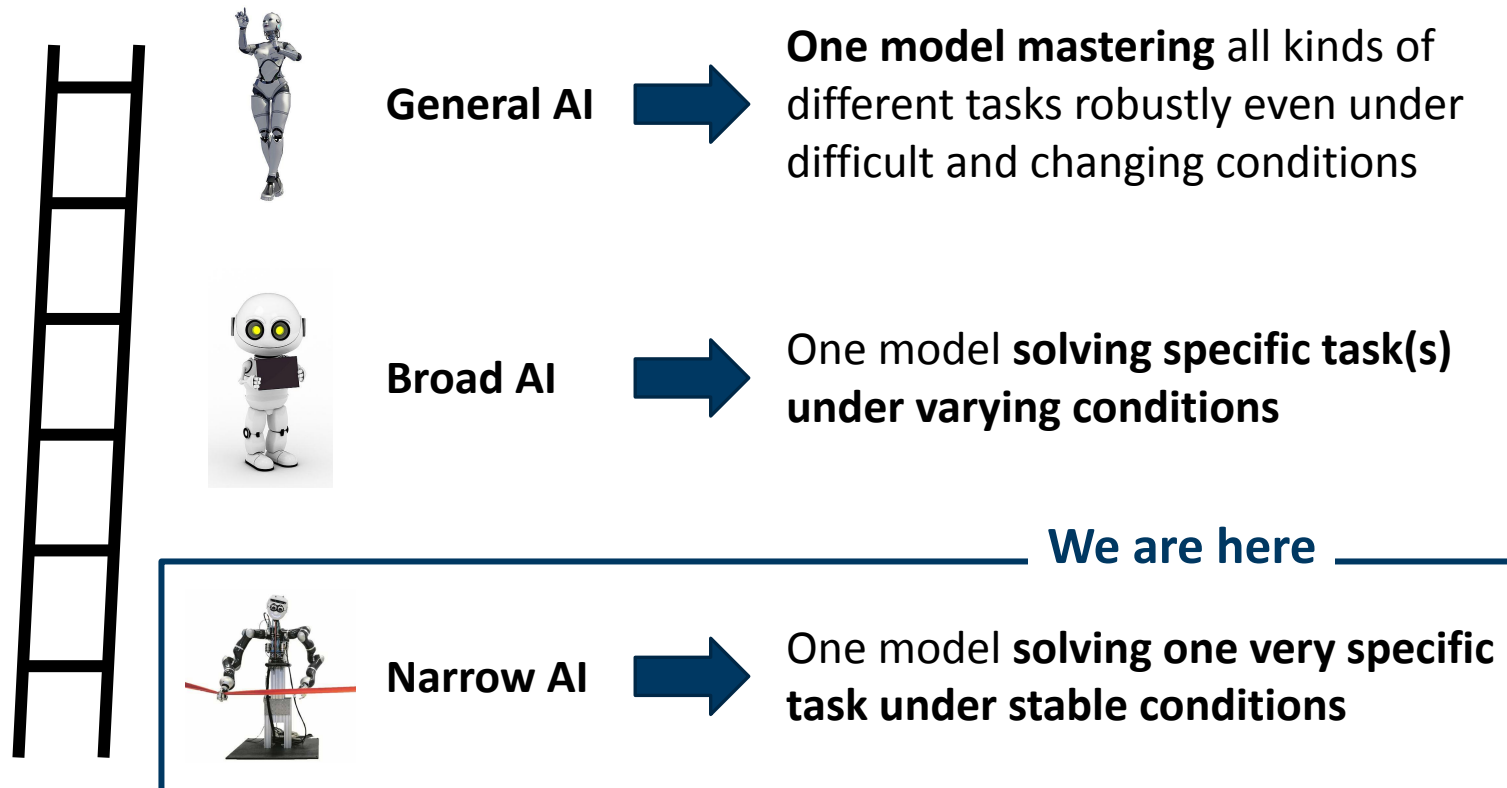
(c) Texture-shape cue conflict

63.9%	Indian elephant
26.4%	indri
9.6%	black swan

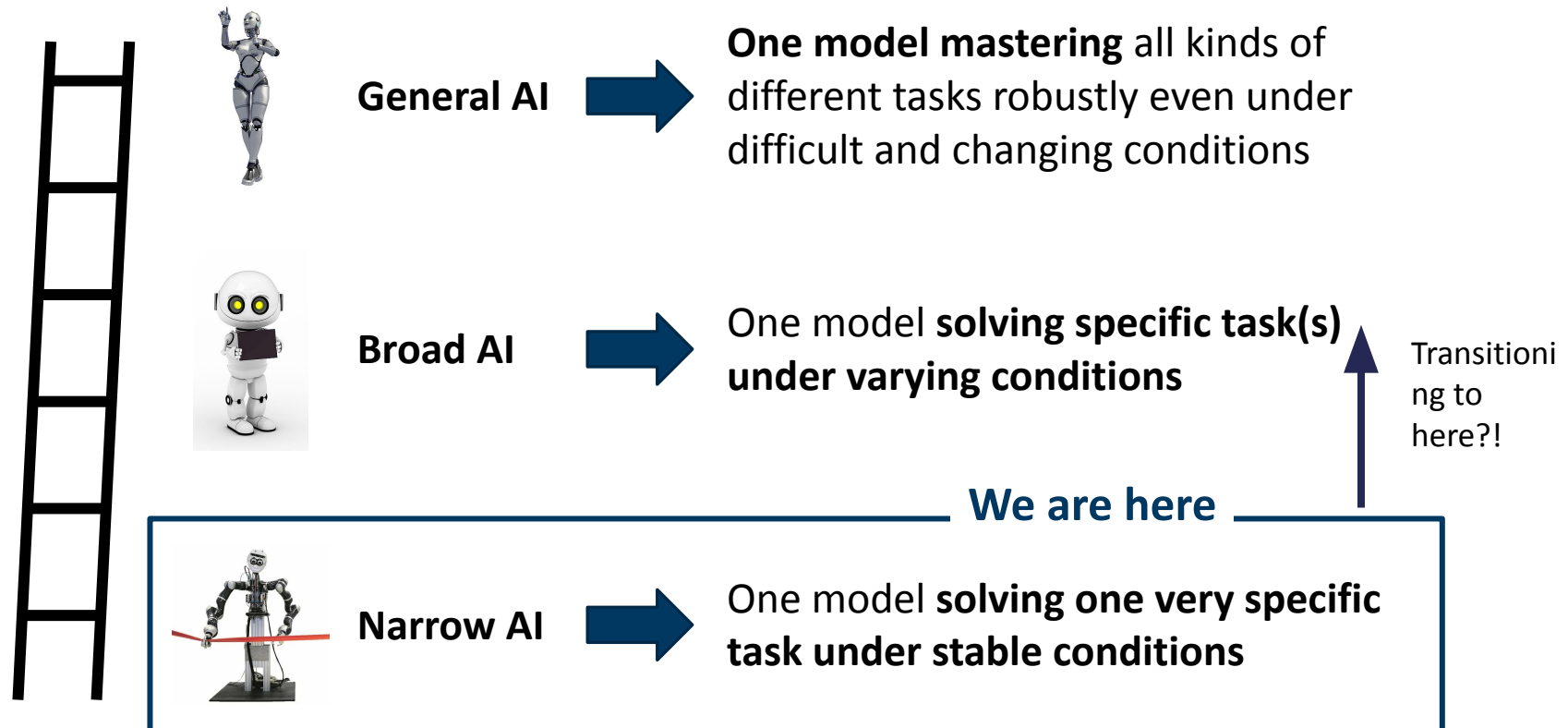
Human vs. neural net: shape or texture?



Summary: Where do we stand?



Summary: Where do we stand?



Questions?
